

Title: Single cell multi-omics enables high-resolution identification and functional purification of human acute myeloid leukemia stem cells

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Key Points

- Single cell multi-omics identifies human AML LSCs at high resolution.
- *HOPX* and *SOCS2* co-expression (hrLSC2) defines a prognostic gene signature in de novo acute myeloid leukemia.
- hrLSC2 marks an AML subpopulation (iLSCs) with a distinct immunophenotype.
- iLSCs can be purified using flow cytometry and are significantly enriched for LSCs.

Summary

In human acute myeloid leukemia (AML), a sub-population of leukemia stem cells (LSCs) drive disease initiation, therapeutic resistance, and relapse. However, the lack of reliable markers to distinguish LSCs from bulk leukemia cells has impeded progress in studying LSC pathogenesis and developing meaningful LSC-specific diagnostics and therapeutics. Existing LSC gene signatures, derived from bulk populations, cannot definitively identify LSCs at single-cell resolution. To address this, we analyzed large patient cohorts with bulk gene expression data and single-cell multi-omic assays to identify a prognostic gene signature that is specifically enriched in a clinically adverse AML sub-population. Using this signature, we defined and prospectively isolated CD34+CD90-CLL1-CD69+CD53- immunophenotypic LSCs that are significantly enriched for LSC content based on limiting dilution xenotransplantation assays. Our findings demonstrate the power of single-cell multi-omics to precisely identify a clinically relevant LSC population and establish a clear framework for future translational research in AML.

Introduction

Acute myeloid leukemia (AML) is an aggressive bone marrow cancer that is associated with a five year survival rate of less than 32%¹. Most patients are either refractory to treatment or relapse with incurable disease². Although current risk stratification systems can provide guidance on treatment^{3,4}, they rely primarily on genetic factors which may incompletely characterize AML severity⁵. The cancer stem cell model can provide additional prognostic insights by describing the functional heterogeneity of the disease cells⁶. According to the cancer stem cell model, AML is organized as a cellular hierarchy that develops from the leukemic transformation of hematopoietic stem and progenitor cells (HSPCs) into leukemia stem cells (LSCs)^{5,7}. LSCs are defined as cells

capable of transplanting leukemia into immunodeficient mice and higher LSC frequency is associated with treatment resistance and disease relapse^{6,7}. Similarly, high LSC content determined by gene expression signatures derived from LSC-enriched cellular fractions of patient samples are associated with poor clinical outcomes⁸⁻¹⁰. These observations suggest that LSCs may need to be eliminated to cure patients. A significant limitation of these prior studies is that the corresponding LSC gene signatures were derived from populations of leukemic cells enriched for LSCs, where the frequency of functional LSCs is variable and heterogeneous. Importantly, it is unclear if these signatures define a unique LSC population or a coordinated gene expression program across a mixture of cells. Therefore, AML LSCs remain ill-defined at the single cell level, which has impaired our ability to study their biological and clinical relevance and advance LSC-targeted therapies.

Effectively addressing these limitations requires the prospective isolation of LSCs at high purity using surface markers. Although there has been a long history of defining LSC-enriched sub-populations in this way¹¹⁻¹⁶, a highly specific immunophenotype remains elusive as it varies significantly from patient to patient. Initially, AML LSC sub-populations were identified in the CD34+CD38- compartment through xenotransplantation assays¹⁷, and were reported to lack the canonical hematopoietic stem cell (HSC) marker CD90¹². Subsequent studies have demonstrated that LSCs are present in both CD34+CD38+ and CD34- fractions at lower frequencies^{9-11,18,19}. Over the following years, several groups have identified additional candidate LSC antigens including TIM3^{15,20}, CD99¹⁶, CD123^{21,22}, CD44²³, CLL1²⁴, CD25²⁵, CD96²⁶, CD47¹⁴, GPR56²⁷, CD93²⁸, CD244²⁹, S1PR3³⁰, and CD117²⁹. Despite these extensive analyses, highly specific AML LSC markers have not been identified. Part of the difficulty is due to different standards for defining LSCs as only a few studies performed limiting dilution assays (LDAs) to quantify LSC content, and different xenograft models can influence transplantation efficiency and stem cell quantification³¹⁻³³. Amongst the xenotransplantation studies that performed LDAs, Eppert et al. observed an LSC frequency between 1/1,600–1/1,100,000 cells in CD34+CD38- cells from 16 primary samples⁹, Xie et al. observed an LSC frequency of 1/882–1/33,314 in CD34+CD33+S1PR3low/- cells from 3 primary samples³⁰, and Pabst et al. observed an LSC frequency of 1/519 in CD34+GPR56+ albeit in a single AML case²⁷. The lack of a specific and generalizable LSC immunophenotype and inability to purify cells at the single cell level has impaired our ability to directly study AML LSC pathogenesis and connect LSC biology broadly with clinical outcomes. Specifically, it is not known if prognostic LSC gene signatures are highly expressed in single cells, if these cells express a unique immunophenotype, and if that immunophenotype can enrich for functional LSCs.

Here, we systematically address these limitations by basing our analysis on four fundamental principles (Figure 1A): 1) LSC burden is associated with worse clinical outcomes, 2) LSC signatures should be expressed in a cell type specific manner, 3) these cells can be purified by flow cytometry, and 4) LSCs should demonstrate a higher capacity to engraft leukemia in immunodeficient mice compared to bulk AML cell populations. Using these principles, we derived a two gene high risk LSC gene expression signature (i.e. hrLSC2) that is both strongly prognostic and highly expressed in specific AML cells. Cells with increased expression of hrLSC2 were defined as candidate LSCs (cLSCs) and were enriched in a population of CD34+CD90-CLL1-CD69+CD53- cells, i.e. immunophenotypic LSCs (iLSCs), based on concurrent scRNA-seq and surface marker quantification. iLSCs were prospectively isolated by Fluorescence-Activated Cell Sorting (FACS) and contained significantly greater LSC frequency compared to bulk CD34+ cells. Overall, our findings provide updated computational and flow cytometry-based strategies for isolating prognostically adverse LSCs in adult human AML.

Definitions (Figure 1B)

1. hrLSC2 (high risk two gene LSC signature): LSC gene expression signature comprised of *HOPX* and *SOCS2*
2. cLSC (candidate LSC): AML cells annotated as HSCs using a normal HSPC-specific classifier that also highly express hrLSC2
3. iLSC (immunophenotypic LSC): CD34+CD90-CLL1-CD69+CD53- AML cells that express high hrLSC2 and are enriched for cLSCs and functional LSCs

Results

hrLSC2 is a simple yet highly prognostic LSC signature in de novo AML

AML risk stratification systems guide clinicians on the diagnosis, prognostication, and management of patients and rely primarily on molecular and cytogenetic profiling^{3,4}. Using these systems, clinicians can categorize AML patients as either favorable, intermediate, or adverse risk which can predict prognosis and influence treatment. Most AML arises de novo, i.e. without prior exposure to chemoradiation or antecedent hematologic disorders and are categorized as intermediate risk. Within this majority, patients have highly variable responses to treatment and clinical outcomes. This highlights a significant limitation of current risk models which may benefit from including features of the AML hierarchy such as LSC content⁵. Bulk gene expression has been shown to correlate with LSC content; however, numerous studies have subsequently generated data that could refine the initial LSC signatures. We therefore revisited the task of identifying an LSC specific and prognostically relevant signature, by curating a large AML cohort ($n=1,949$) of published bulk gene expression data with survival annotations³⁴⁻⁴⁰. We focused our analysis on de novo AML evaluated at diagnosis and excluded acute promyelocytic leukemia which can be considered a separate AML subtype as it has a distinct treatment paradigm and highly favorable outcomes. This resulted in a core set of 942 cases (Table ST1, Figure S1). We then narrowed the gene expression search space to include genes that have been associated with LSC biology (Table ST2)^{8-10,41}. After data normalization and scaling, the cohort was split into training ($n=626$) and test ($n=316$) sets which were well balanced (Figure S2). We subsequently evaluated the training dataset using ensemble machine learning methods to identify genes that were most associated with shorter survival. We aggregated the ranked lists (Table ST3) generated from these ensemble methods to identify the top features⁴², and observed that higher expression of *HOPX* and *SOCS2* were the most important features associated with worse survival in both de novo and normal karyotype AML, which was used as a surrogate for intermediate risk AML (Table ST4).

We then generated a weighted score based on *HOPX* and *SOCS2* expression, i.e. hrLSC2 (Figure 1B), using the training dataset and evaluated its performance for predicting survival in the test dataset by conducting iterative cox proportional-hazards (coxph) regressions ($n=1,000$; Figure 1C). hrLSC2 and LSC17¹⁰, a rigorously validated LSC signature, performed better than other gene signatures⁴³⁻⁴⁵ in de novo AML (Figure 1C, left panel). When evaluating normal karyotype AML, hrLSC2 was superior to other LSC signatures (Figure 1C, right panel). The improved performance of hrLSC2 was highlighted further in a multi-variate coxph analysis including other LSC signatures, age, sex, *FLT3*-ITD status (high risk mutation), *NPM1* mutation status (low risk mutation), and cytogenetic risk category (Figure 1D; $n=176$ test cases containing both mutation and cytogenetic annotations). Compared to all other covariates, hrLSC2 was most significantly associated with worse survival in de novo AML (Figure 1E) and further stratifies survival in conjunction with established risk features (Figure 1F-H). Overall, this analysis supports that hrLSC2 is highly prognostic and may improve patient risk stratification when incorporated with genomic features.

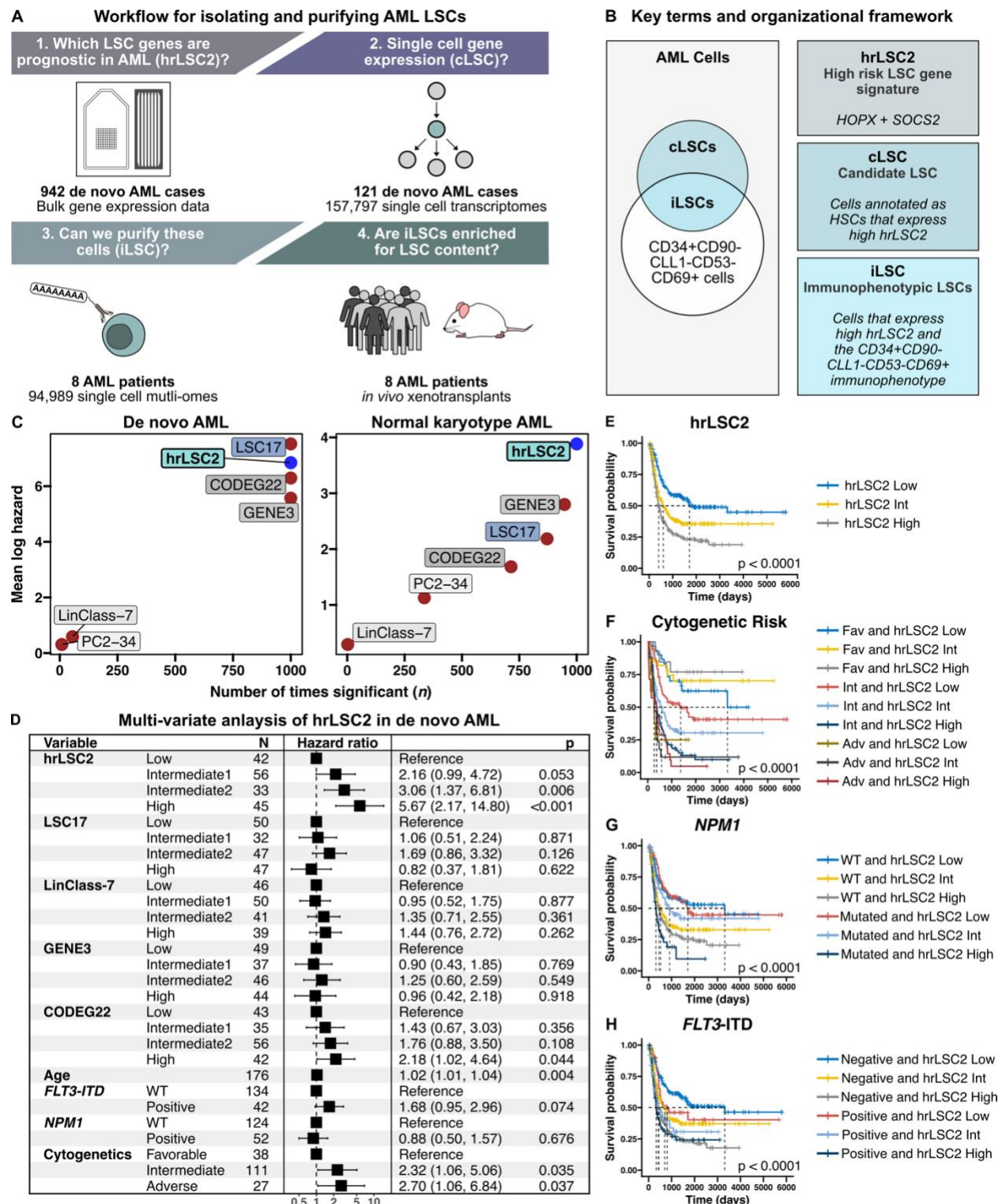


Figure 1: hrLSC2 is a prognostic gene signature in de novo AML. A) General workflow for identifying an LSC-specific gene signature in de novo AML. B) Key terminology used throughout the manuscript and Venn diagram illustrating relevant LSC sup-populations evaluated in this study. C) Results of iterative coxph regression analysis

($n=1000$) in de novo and normal karyotype AML evaluating the performance of hrLSC2 against previously published prognostic AML signatures for predicting survival. D) Multi-variate coxph survival analysis evaluating the association between prognostic AML signatures, age, high and low risk mutation status, and cytogenetic risk categories. E-H) Kaplan-Meier survival analysis evaluating hrLSC2 status in combination with established high and low risk features. Statistical significance was tested using either the Wald or Log-rank test.

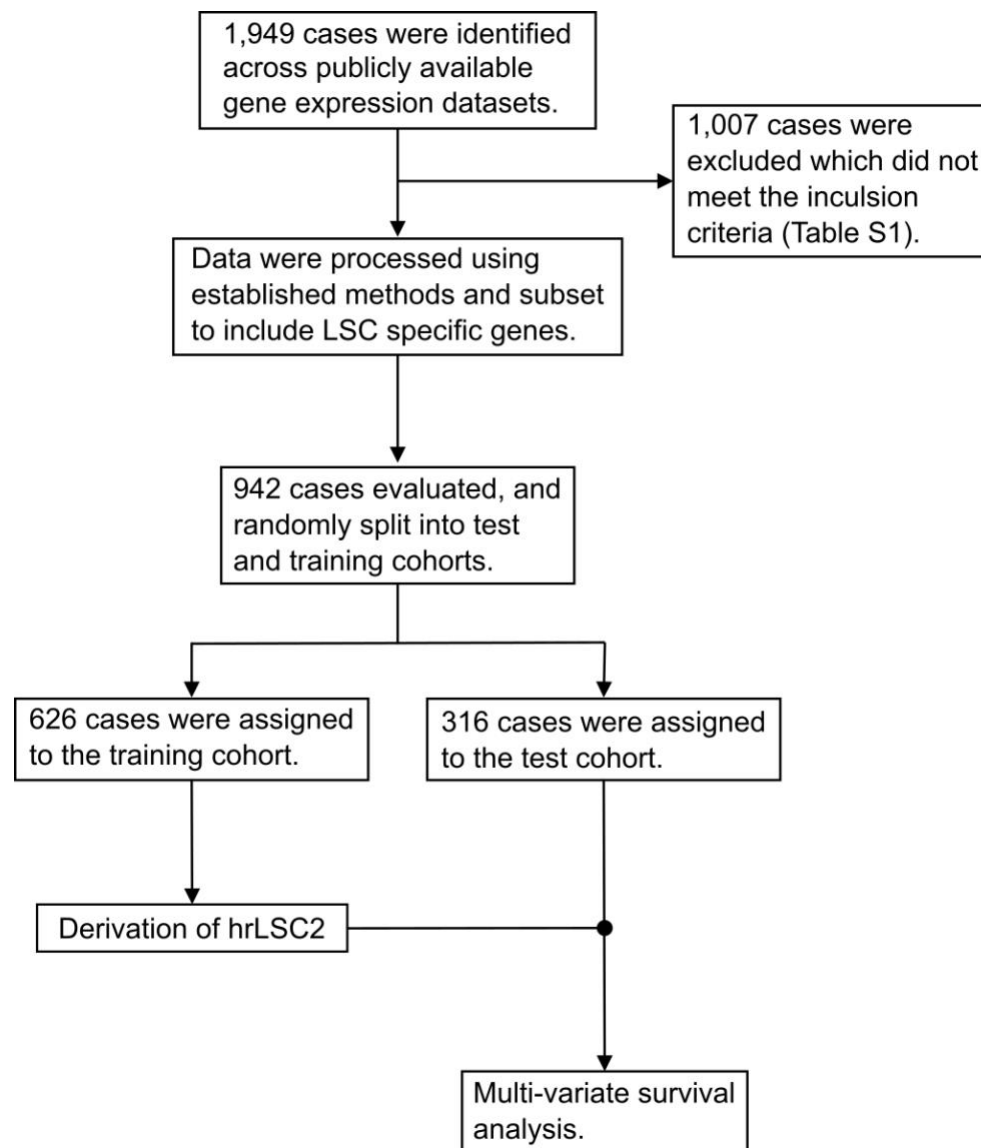


Figure S1: Dataset curation, filtration, and analysis workflow for deriving hrLSC2.

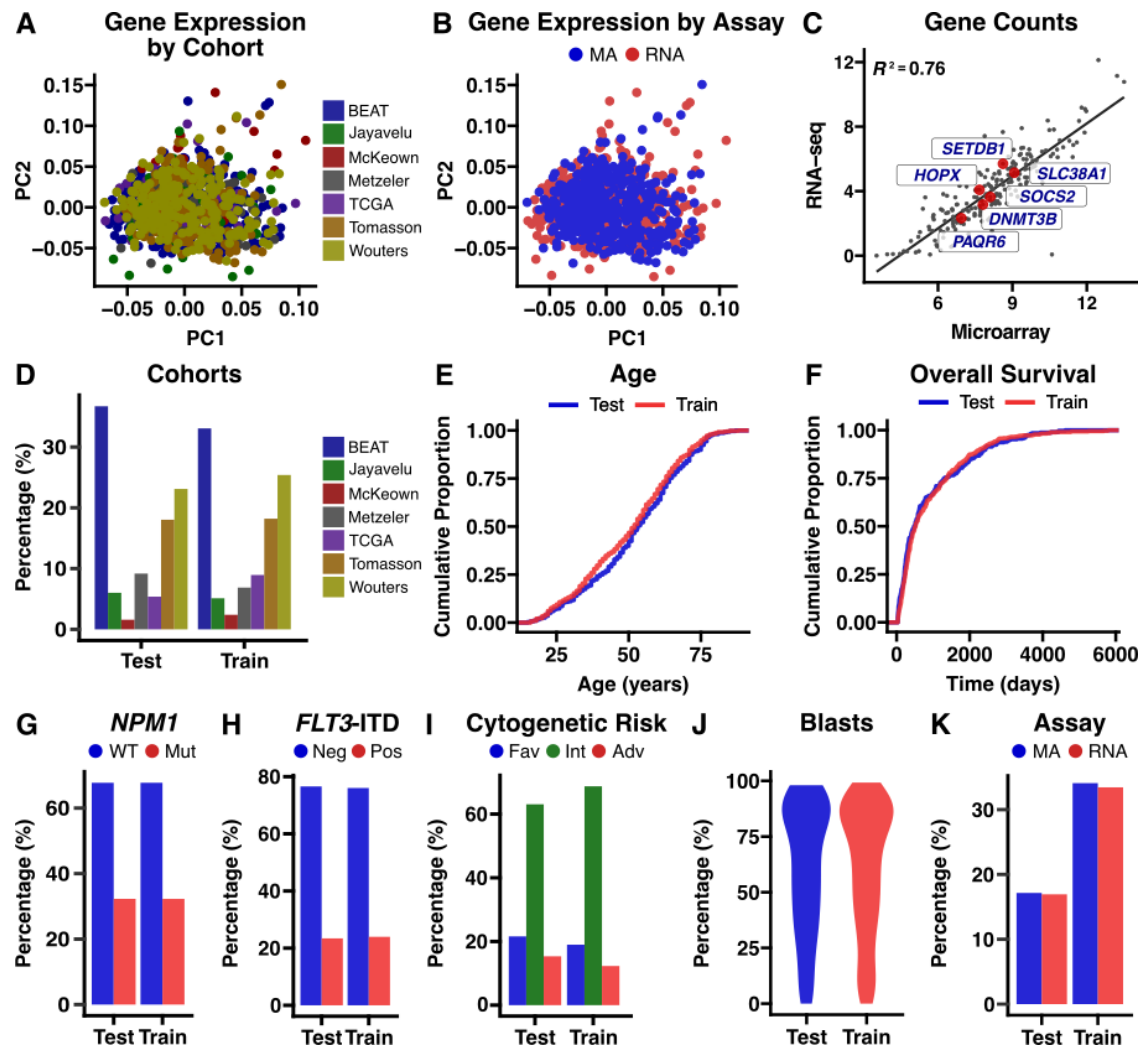


Figure S2: Description of key dataset metrics. Principal component plot of gene expression labeled by (A) cohort and (B) assay subtype. C) Correlation between processed gene counts from RNA-seq and MA data with highly prognostic genes labeled. D) Bar plot showing relative proportions of each cohort in the training and test datasets. Comparison of training and test dataset compositions for key characteristics including (E) age, (F) survival, (G-J) clinical features, and (K) assay. MA – microarray, Neg – negative, Pos – positive, RNA – RNA-seq.

High hrLSC2 expression marks a prognostically relevant sub-population in de novo AML

We subsequently evaluated whether hrLSC2 expression could allow us to specifically identify a highly expressing cellular sub-population across a large cohort of de novo AML. To address this goal, we compiled publicly and internally generated scRNA-seq data ($n=985,422$ cells) from patients with de novo AML ($n=121$) and healthy donors ($n=25$) (Figure S3, Figure 2A, S4A, and Table ST7). After applying the same processing and quality control methods to each dataset (Figure S3), we removed lymphocytes and annotated cells using an HSPC-specific classifier that we previously developed to accurately annotate human HSPCs in scRNA-seq data⁴⁶. The data ($n=157,797$ cells) were then integrated using scGen (Figure 2B)⁴⁷ and hrLSC2 expression was calculated for each cell. To determine if hrLSC2 could identify an AML sub-population, we first evaluated its expression across all AML cells and observed a multi-modal distribution (Figure S4B). We then used the expectation-maximization algorithm to fit finite mixtures based on hrLSC2 expression⁴⁸, which allowed us to identify cells that express hrLSC2 at high levels. These cells were enriched for cells mapping to the most primitive progenitors, including HSCs and LMPP1s (Figure S4C) and were more homogenous based on lower diversity indices (Figure S4D) and cellular spread (Figure S4E) compared to all cell types for each donor (Methods). The data were down sampled to ensure an equivalent number of cells were being compared to the hrLSC2 high sub-population. As expected, cells annotated as HSCs and LMPP1s not only expressed hrLSC2 at high levels (Figure 2C) but also contained a high fraction of cells that co-expressed *HOPX* and *SOCS2* compared to all other cell types (Figure 2C-D). These observations support that high hrLSC2 derived from both genes, rather than high levels from one gene only, is important for marking these cells. To computationally enrich for hrLSC2-expressing cells within the AML HSPC compartment, we applied an expectation-maximization algorithm to identify cells exhibiting high hrLSC2 expression. Since AML-HSCs demonstrated the highest hrLSC2 expression levels among all cellular subsets (Figure 2C) and is considered the most primitive cell type, we focused the subsequent analysis on this population. AML-HSCs expressing high hrLSC2 were designated as candidate LSCs (cLSCs) (Figure 2E) and showed significant enrichment for cells co-expressing *HOPX* and *SOCS2* relative to AML-HSCs with lower hrLSC2 expression. AML-HSCs exhibiting low hrLSC2 expression retained their original cellular annotation as an AML-HSC (Figure 2F-G).

Using the updated annotations, the data were re-integrated using scGen (Figure 2H, S4F), which was then used to construct a signature matrix for bulk gene expression deconvolution using CIBERSORTx⁴⁹. The data were split into training and test datasets, and the signature matrix was constructed using the training data. To evaluate deconvolution performance, we generated artificial bulk transcriptomes using either a single population (pure) or mixture of cell types (synthetic mixtures) from single cells in the test dataset. We observed robust deconvolution performance using our high-resolution signature matrix ($n=32$ cell types) on mixtures derived from the test dataset (Figure S4G-H) which motivated us to perform a large-scale deconvolution analysis on our cohort of de novo AML (Table ST1). The deconvolution results show balanced cell type imputations across each cohort (Figure S4I). We then evaluated the association between cell type frequencies and survival using ensemble machine learning methods (Table ST8) and observed that cLSC content was the most important feature for predicting survival. Importantly, other cell types including hrLSC2 high LMPP1s (cLSC-LMPP1) were weakly associated with outcome, highlighting that subsetting on the most immature cell subset derived from scRNA-seq, i.e. the HSC, increases prognostic power. The strength of this association is illustrated in Figure 2I which presents the results from iterative coxph regressions ($n=1,000$) for each cell type, and Figure 2J which shows how cLSC content can effectively stratify patient survival outcomes.

We further evaluated the clinical relevance of cLSCs by performing a multi-variate Cox analysis which included LSC signatures, age, sex, *FLT3*-ITD status, *NPM1* mutation status, and

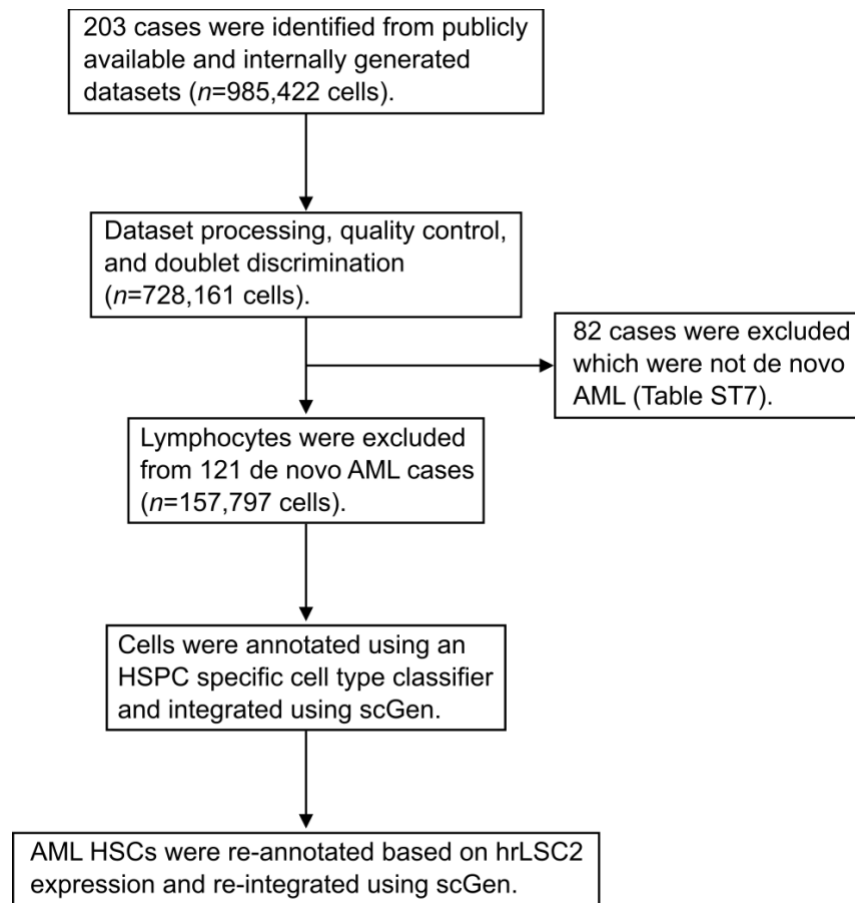


Figure S3: Single cell RNA-seq dataset curation, processing, annotation, and integration workflow.

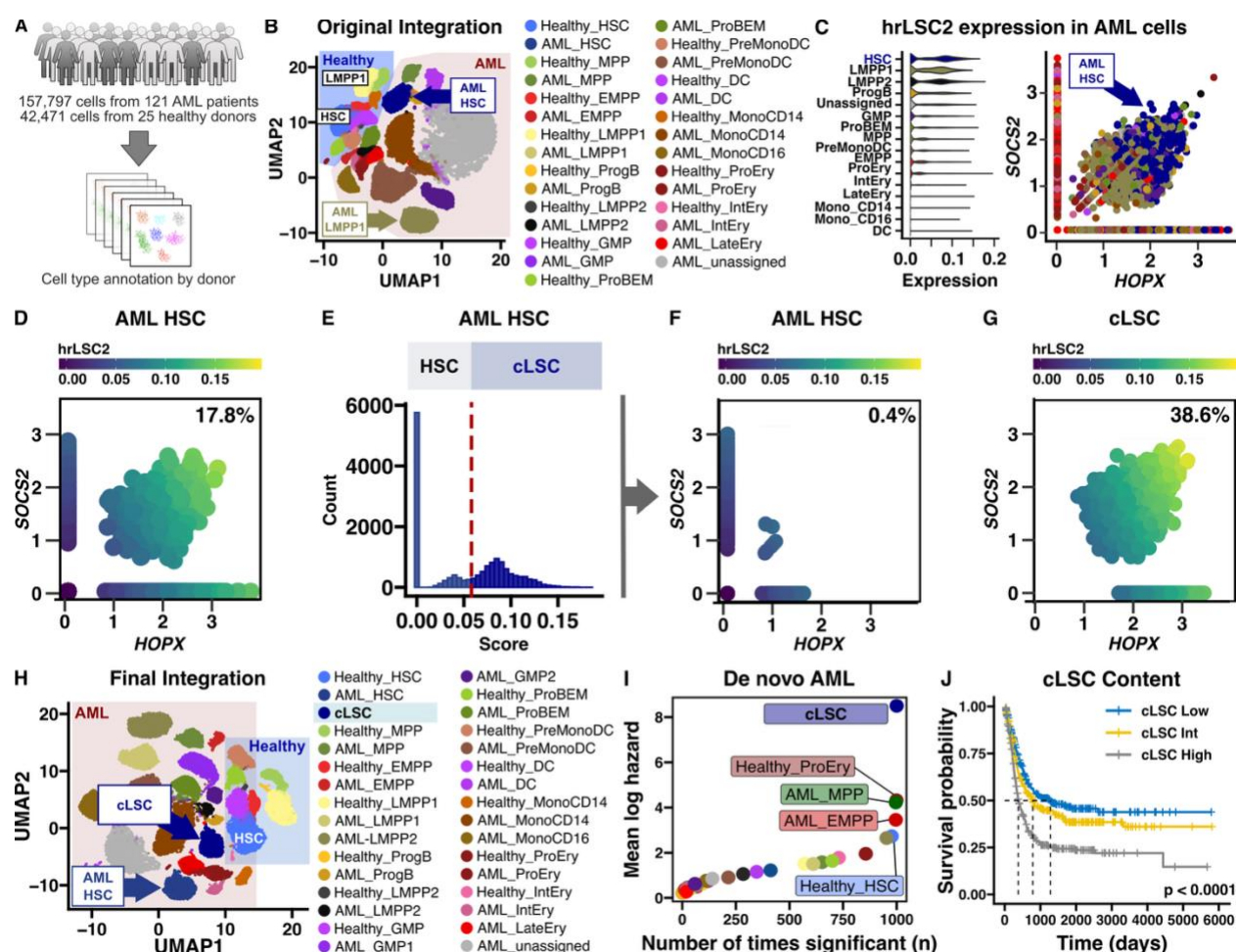


Figure 2: cLSCs are the most prognostic characterized cell type in de novo AML.

A) Workflow for constructing an integrated scRNA-seq dataset for de novo AML. B) Post integration UMAP projection of cells from AML patients and healthy donors annotated using an HSPC-specific classifier. C) hrLSC2 expression in AML cells based on normal cell type annotations. D) Expression of *HOPX* and *SOCS2* for AML cells annotated as HSCs. Percentage reflects cells that co-express *HOPX* and *SOCS2*. E) Histogram illustrating hrLSC2 expression for AML cells annotated as HSCs. Dashed red line indicates threshold for annotating cLSCs. F-G) Expression of *HOPX* and *SOCS2* for cells re-annotated based on hrLSC2 expression. H) UMAP projection of the final integration after updating cell type annotations for hrLSC2 expression. I) Results of iterative Cox regression analysis ($n=1000$) in de novo AML evaluating the performance of cell type content for predicting survival. J) Kaplan-Meier survival analysis evaluating cLSC content. Statistical significance was tested using either the Wald or Log-rank test.

A Workflow for identifying prognostic cell types in de novo AML

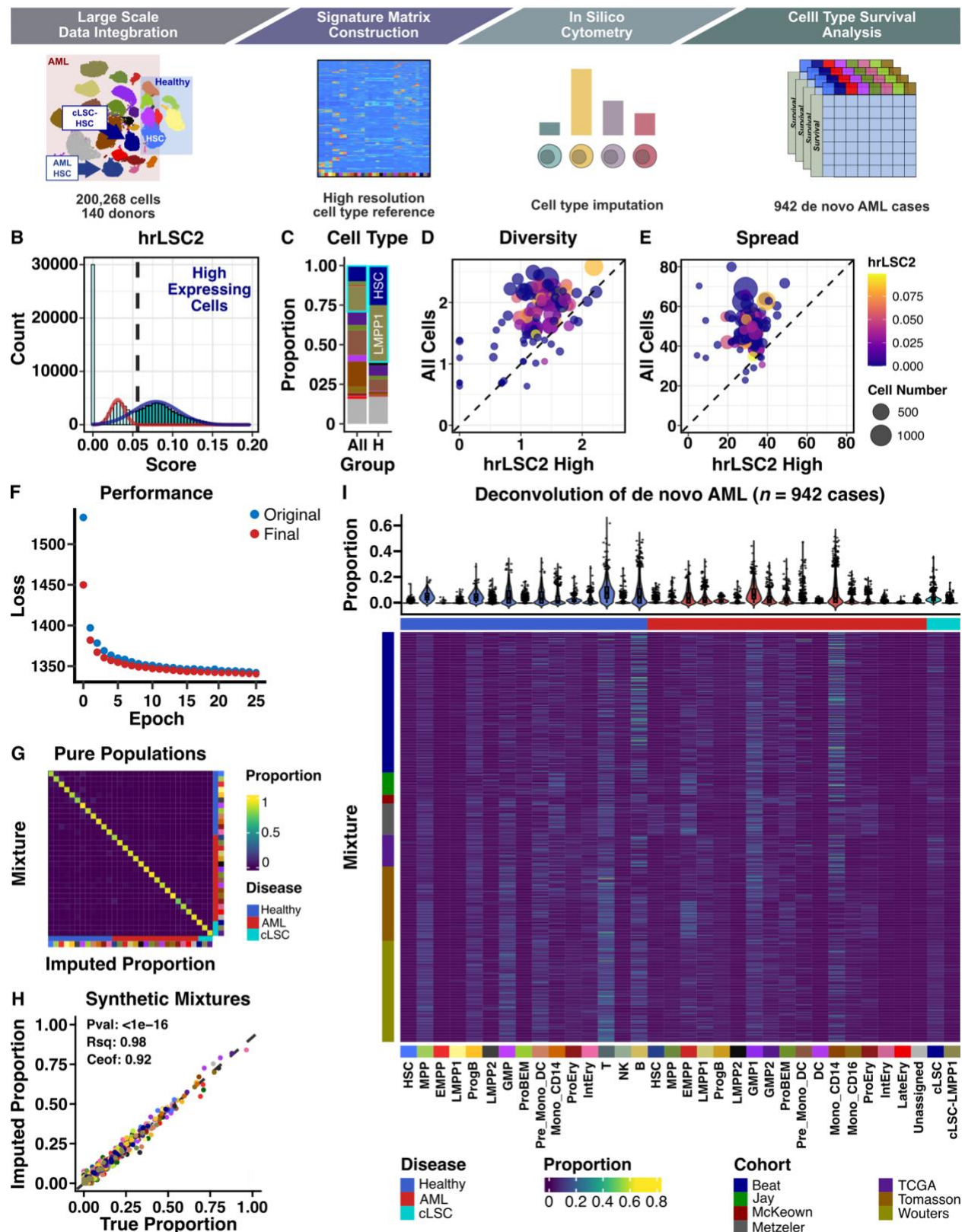


Figure S4: Constructing a high-resolution single cell reference for bulk gene expression deconvolution. A) Workflow for constructing a high-resolution signature matrix for a cell type specific survival analysis in de novo AML. B) Histogram of hrLSC2 expression across all AML cells. Dashed blue line indicates threshold for identifying cells expressing high levels of hrLSC2. C) Cell type frequencies within all cells and the hrLSC2 high (H) sub-population. Comparison of (D) Shannon diversity indices and (E) spread by sample for the hrLSC2 high sub-population and all cell types. The comparison containing all cells was down sampled to ensure the same number of cells. Each dot represents a single donor and is colored based on median hrLSC2 expression and sized based on cell number. F) Loss comparison over training epochs for both the original and final integration with scGen. G) Heatmap illustrating imputation results from deconvolving artificial bulk transcriptomes derived only from cells in each respective cluster. H) Comparison between imputation results and true cell type proportions from deconvolving artificial bulk transcriptomes containing mixtures of cell types at known frequencies. I) Deconvolution results of 942 de novo AML cases using CIBERSORTx. The violin plot presents cell type frequencies, and the heatmap is colored based on cohort (rows) and disease (columns).

cytogenetic risk category (Figure 3A). We find that cLSC content is significantly associated with worse outcomes and interacts with mutation status and cytogenetics for predicting survival (Figure 3B-D). Since shorter survival in AML is often related to relapsed disease, we next evaluated changes in cLSC content at relapse. To address this goal, we compiled an AML gene expression cohort containing paired cases from diagnosis and relapse after either cytotoxic chemotherapy ($n=47$ cases) or allogeneic hematopoietic cell transplant (HCT; $n=26$ cases)^{36,50-54}. Patients who receive chemotherapy alone are generally considered to have lower risk disease compared to those who receive HCT⁵⁵. After imputing cell type frequencies using CIBERSORTx, we observed a significant increase in cLSC content at relapse post chemotherapy (Figure 3E). Although we do not see an increase in cLSC content after HCT (Figure 3F), there is significantly greater cLSC content at diagnosis in AML cases that receive HCT compared to those that receive chemotherapy (Figure 3G), and no difference in cLSC content after relapse comparing either treatment modality (Figure 3H). To further characterize the relevance of cLSCs after treatment, we compared changes in cLSC frequency to those of all other cell types (Figure 3I). For patients receiving chemotherapy, the change in cLSC content was not only the most significant but was also associated with the largest effect size (Cohen's $d=0.85$). In contrast to cLSCs, the other prognostic cell types identified by survival analysis (Table ST8) showed minimal changes at relapse (Figure 3J). Overall, these findings demonstrate that cLSCs are both the highest hrLSC2 expressing cells and the most prognostic cell type in de novo AML.

Cells that express hrLSC2 at high levels can be identified using a unique immunophenotype

Since hrLSC2 was derived from gene sets associated with LSC content (Figure 2C), we hypothesized that cLSCs are bona fide, functional LSCs as they express the highest level of hrLSC2. To test this hypothesis, we first evaluated if these cells also express a unique immunophenotype with which they could be prospectively isolated with FACS. For this purpose, we performed scRNA-seq using a targeted mRNA panel ($n=500$) and scADT-seq ($n=39$) on 142,628 cells across 8 primary adult AML samples (Table ST5; Figure 4A). Since LSCs are often a rare sub-population, we performed a targeted mRNA analysis to efficiently profile more cells per sample compared to a whole transcriptome analysis and therefore used cells with high expression of hrLSC2 as a surrogate for cLSCs. After strict quality control, doublet discrimination, and lymphocyte, erythrocyte, and monocyte removal (Table ST6)⁴⁶, we identified 94,989 high quality cells for subsequent analysis. hrLSC2 scores were derived per cell based on expression of *HOPX* and *SOCS2* and cells above the seventy-fifth percentile were marked as cLSCs (Figure S5). The threshold of seventy-five percent was determined based on balancing the goal of identifying cells expressing the highest level of hrLSC2 while maintaining the power to detect important features (Figure S5E). We then performed a Least Absolute Shrinkage and Selection Operator (Lasso) classification comparing ADT expression to cLSC cell status for each sample to identify protein markers compatible with FACS⁵⁶. We observed that CD69, CD53, CD34, and CLL1 were the most important features associated with cLSC status (Figure 4B). Using the relative importance of CD69 and CD53 compared to CD34 and CLL1 as a guide, we created an iterative gating strategy for purifying cLSCs (Figure 4C-E). Using this approach, we computationally isolated a population of CD34+CLL1-CD90-CD69+CD53- cells with significantly higher hrLSC2 expression (Figure 4F) and designated this immunophenotypically defined subpopulation as iLSCs. These cells also expressed higher levels of LSC17¹⁰ and LinClass7⁴³, two previously published LSC signatures, compared to other sub-populations⁵⁷, albeit at a smaller fold difference than hrLSC2 (Figure 4G-H). The final CD34+CLL1-CD90-CD69+CD53- cell population was also more homogenous (Figure 4I) compared to cells purified using previously validated LSC sorting strategies such as CD34+CD38- and expression of CD99 and TIM3⁵⁷ (Figures 4J-L).

To further validate our findings, we performed a similar analysis using the updated sorting strategy on 4 cases with concurrent whole transcriptome quantification and scADT-seq data (Table ST5)

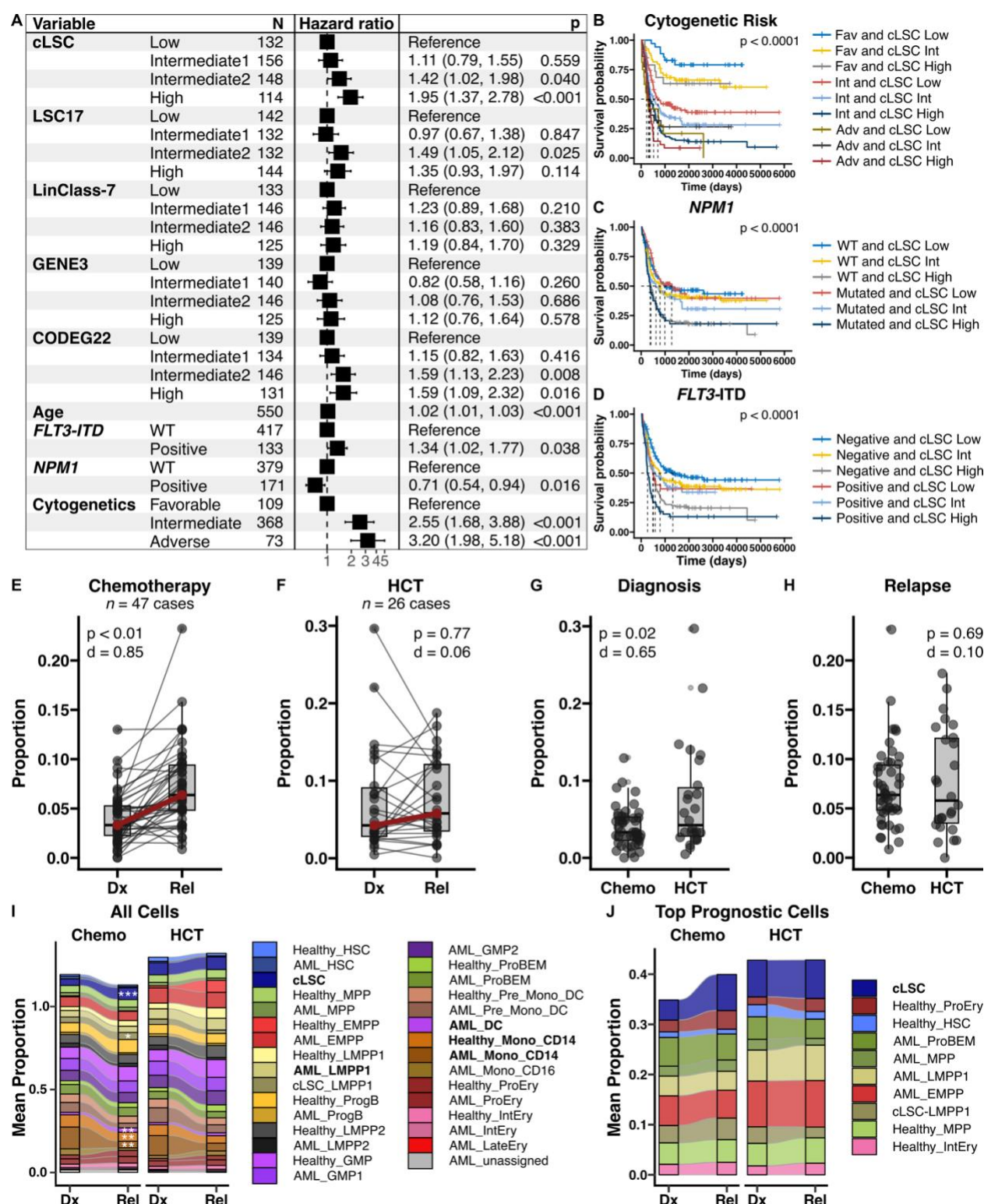


Figure 3: cLSCs are associated with worse outcomes in de novo AML and expand in relapsed disease. A) Multi-variate coxph survival analysis evaluating the association between cLSC content, prognostic AML signatures, age, high and low risk mutation status, and cytogenetic risk categories. B-D) Kaplan-Meier survival analysis evaluating

cLSC content in combination with established high and low risk features. cLSC content at diagnosis and relapse after (E) chemotherapy and (F) HCT. cLSC content for patients receiving chemotherapy or HCT at (G) diagnosis and (H) relapse. Red line highlights median values. P-values and effect sizes (Cohen's *d*) are indicated. I) Mean cell type proportions at diagnosis and relapse for patients receiving either chemotherapy or HCT. J) Mean proportion of top prognostic cell populations at diagnosis and relapse for patients receiving either chemotherapy or HCT. Statistical significance was tested using either the Wald or Log-rank test for panels A-D. Statistical significance was tested using t-test with multiple hypothesis correction for panels E-J. ***Adjust p-value < 0.0001, **Adjusted p-value < 0.001, *adjusted p-value < 0.05. Chemo – chemotherapy, Dx – diagnosis, HCT – hematopoietic cell transplant, Rel – relapse.

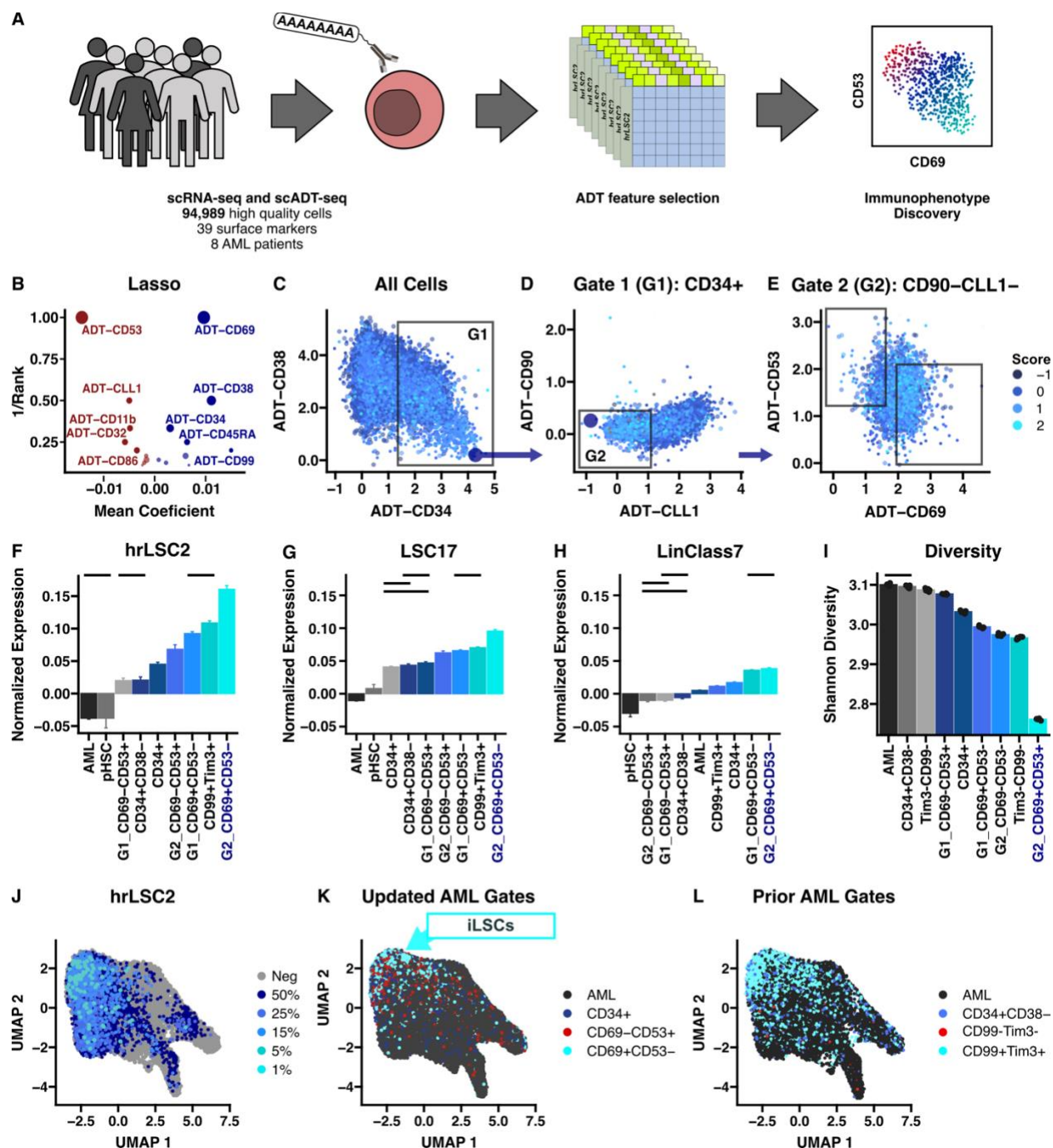


Figure 4: Cells expressing high hrLSC2 also express a unique immunophenotype.

A) Schematic illustrating the workflow for identifying an immunophenotype for cLSCs (i.e. iLSCs) using a targeted gene expression panel and scADT-seq. B) Plot illustrating ADT rank and mean coefficient from Lasso regressions by sample comparing ADT expression and hrLSC2 status. C-E) Computational sorting strategy for purifying iLSCs. Normalized expression of F) hrLSC2, G) LSC17, and (H) LinClass7 within each computationally purified sub-population. I) Diversity measured by the Shannon diversity index for each purified sub-population. All comparisons are statistically significant unless marked by a

solid line. UMAP projections illustrating (J) hrLSC2 expression or sub-population gating based on the (K) updated sort or (L) prior gating strategy. Statistical significance was tested using a Wilcoxon ranked sum test with multiple comparison adjustments with an adjusted p-value < 0.05 considered significant. scADT-seq – single cell antibody directed tag sequencing, G1 – gate 1, G2 – gate 2.

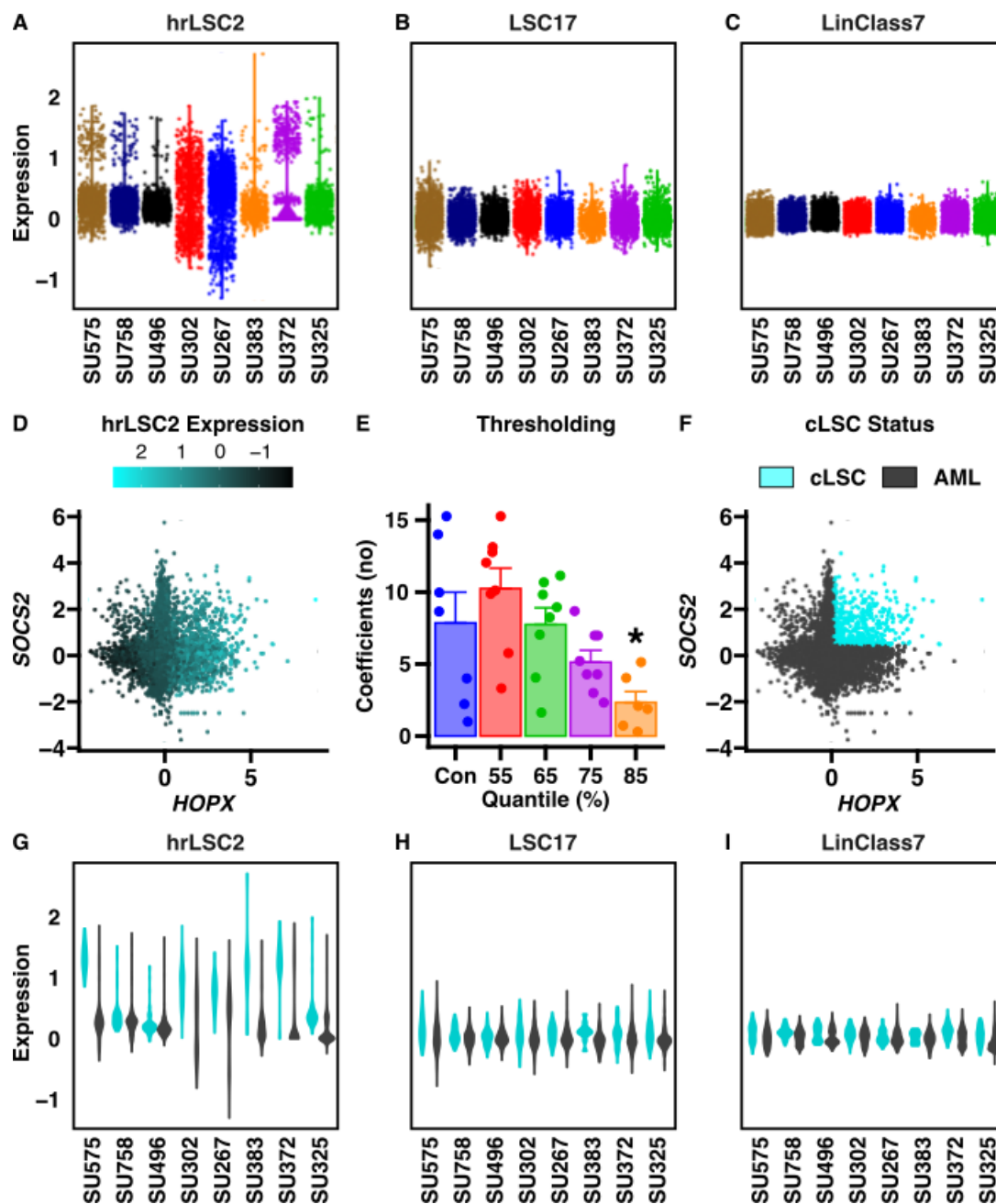


Figure S5: hrLSC2 expression can be used to mark a sub-population of AML cells.

Cellular expression of (A) hrLSC2, (B) LSC17, and (C) LinClass7 for each donor. D) Scatter plot comparing *HOPX* and *SOCS2* expression, where each cell is colored based on its calculated hrLSC2 score. E) Number of non-zero coefficients using Lasso regression comparing ADTs and hrLSC2 as a continuous variable (Con) or using Lasso classification based on quantile thresholding for determining cLSC status. A threshold of 75% (i.e. the top 25% being considered cLSCs) was determined to be optimal as there is a significant decrease in coefficients with higher thresholding. F) Scatter plot comparing

HOPX and *SOCS2* expression and cLSC status based on top 25% thresholding. G-I) Expression of gene signatures based on cLSC status. Statistical significance was tested using a t test or ANOVA with multiple comparison adjustments. *Adjusted $p < 0.05$.

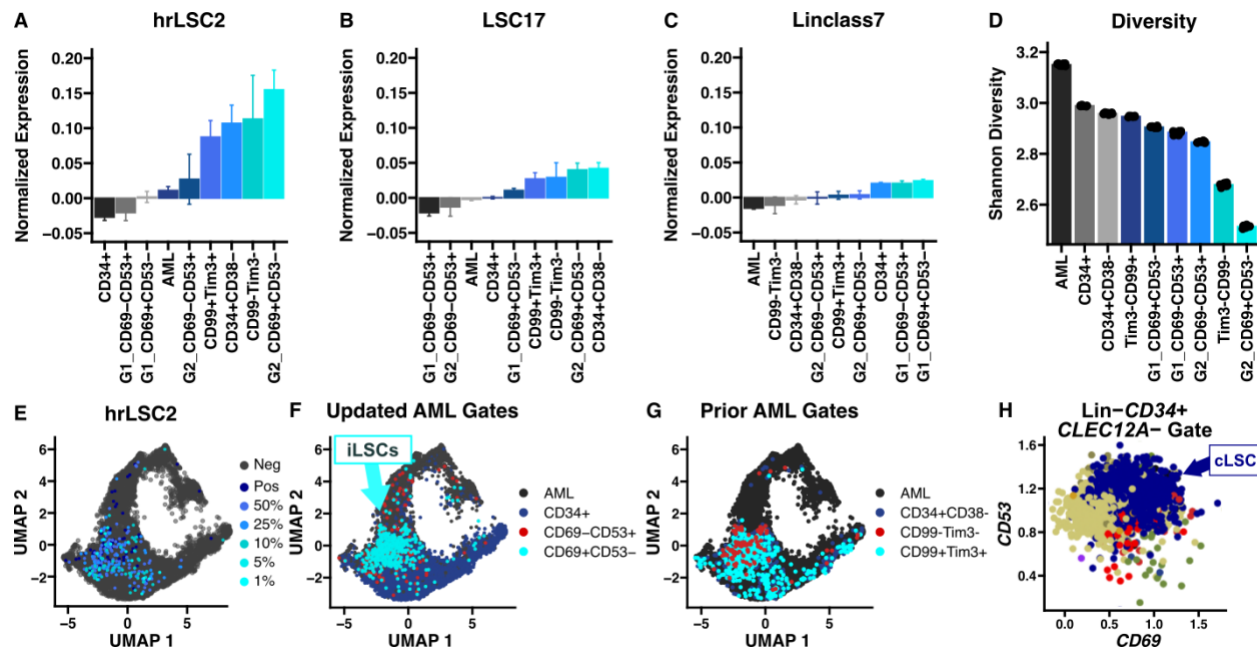


Figure S6: Computationally purifying iLSCs can increase sub-population hrLSC2 expression and purity. Results from purifying AML cells using whole transcriptome scRNA-seq and scADT-seq. Normalized expression of (A) hrLSC2, (B) LSC17, and (C) LinClass7 within each purified sub-population. D) Diversity measured by the Shannon diversity index for each purified sub-population. UMAP projections illustrating (E) hrLSC2 expression or sub-population gating based on the (F) updated computational defined sub-populations or (G) prior LSC gating strategies. H) *CD53* and *CD69* gene expression for cells computationally purified from the scRNA-seq cohort (Figure 2) based on negative lineage marker (Lin-), *CD34*, and *CLEC12A* (gene encoding for CLL1) expression which corresponds to G2. G1 – gate 1, G2 – gate2.

and saw similar patterns in enrichment of gene signature expression (Figure S6A-C) and purity (Figure S6D-G). We then evaluated the broader relevance of this observation by computationally purifying cells from our large scRNA-seq dataset ($n=121$ AML cases; Figure 2) using expression of the genes defining the iLSC immunophenotype, as only RNA data was available for this large number of cases. We note that cLSCs reside within the Lin-CD34+CLEC12A- gate (Figure S6H) which is comparable to the iLSC immunophenotype derived from our multi-omic analysis (Figure 4). These observations suggest that an iterative strategy is most effective for isolating a relatively pure population of cells expressing hrLSC2 at high levels, which we identify as iLSCs.

cLSCs have distinct biomolecular programs compared to AML HSCs and Healthy HSCs

We subsequently evaluated whether cLSCs exhibit unique biological properties compared to other HSCs by performing gene set enrichment analyses (GSEA) on differentially expressed genes between cLSCs and the combination of AML-HSCs and healthy HSCs or cLSCs and healthy HSCs alone (Figure S7). We identified 307 and 265 significant gene ontology (GO) terms respectively which are presented in Figure S7A-B, and the top GO terms are presented in Figure S7C-D. The terms were clustered based on semantic similarity and labeled based on over-represented words present in the GO term descriptions. cLSCs up-regulate pathways associated with cytoskeleton organization, motility, cell activation, differentiation, and apoptosis. We also observe down-regulation of pathways associated with erythroid biology, RNA biology, and proteostasis. The analysis suggests that cLSCs differentially express several key pathways that may influence their development and pathogenesis.

CD34+CLL1-CD90-CD69+CD53- iLSCs are enriched for LSC content

A primary goal of this study is to identify an LSC-enriched population and iLSCs appear to represent such a population as they are associated with worse outcomes in de novo AML, expand in relapsed disease, and express a unique immunophenotype. We therefore evaluated if iLSCs are enriched for LSC content, by designing a FACS-strategy based on our computational observations (Figure S8). Cells from 8 primary AML samples (Table ST5) were purified using this approach and evaluated *in vivo* for LSC content⁵⁸ (Figure 5A). This cohort comprises primarily de novo AML samples that are diverse with respect to age, sex, and mutational status, and harbor common AML-associated mutations including those in *NPM1*, *FLT3*, and *DNMT3A*. We defined LSCs as cells that produce human CD45+HLA+CD33+CD19- engraftment in xenotransplanted mice ($>0.1\%$ human to mouse chimerism) as has been standard in the field⁷. We found that CD34+CLL1- cells engraft at high rates compared to CD34+ cells (Figure 5B). Importantly, we also observed that CD34+CLL1-CD90-CD69+CD53- cells produced significantly greater relative engraftment compared to CD34+ cells and CD34+CLL1-CD90-CD69+CD53+ cells (Figure 5C, Figure S9). We then performed an LDA using primary cells from 3 AML donors (Table ST9) and observed a significantly greater LSC frequency (13 to 162-fold increase) in CD34+CLL1-CD90-CD69+CD53- cells compared to CD34+ cells (Figure 5D-F). Overall, purifying primary AML cells using the iLSC immunophenotype significantly enriches for LSC content across a diverse panel of primary adult human AML samples.

Discussion

Here, we report an updated LSC gene signature, hrLSC2, which is highly prognostic in de novo AML (Figure 1-3) and is significantly enriched in a sub-population of human AML cells with increased LSC content (Figure 5). Our approach for deriving hrLSC2 builds on prior work^{8-10,41} by evaluating a large and diverse cohort of 942 de novo AML cases (excluding acute promyelocytic leukemia; Table ST1) at diagnosis. By focusing the survival analysis on LSC specific genes, we identified the most clinically relevant features associated with LSC biology. The subsequent iterative evaluation using distinct machine learning methods allowed us to confidently identify a small subset of genes for model building (Table ST3-4). We used this information to create a two

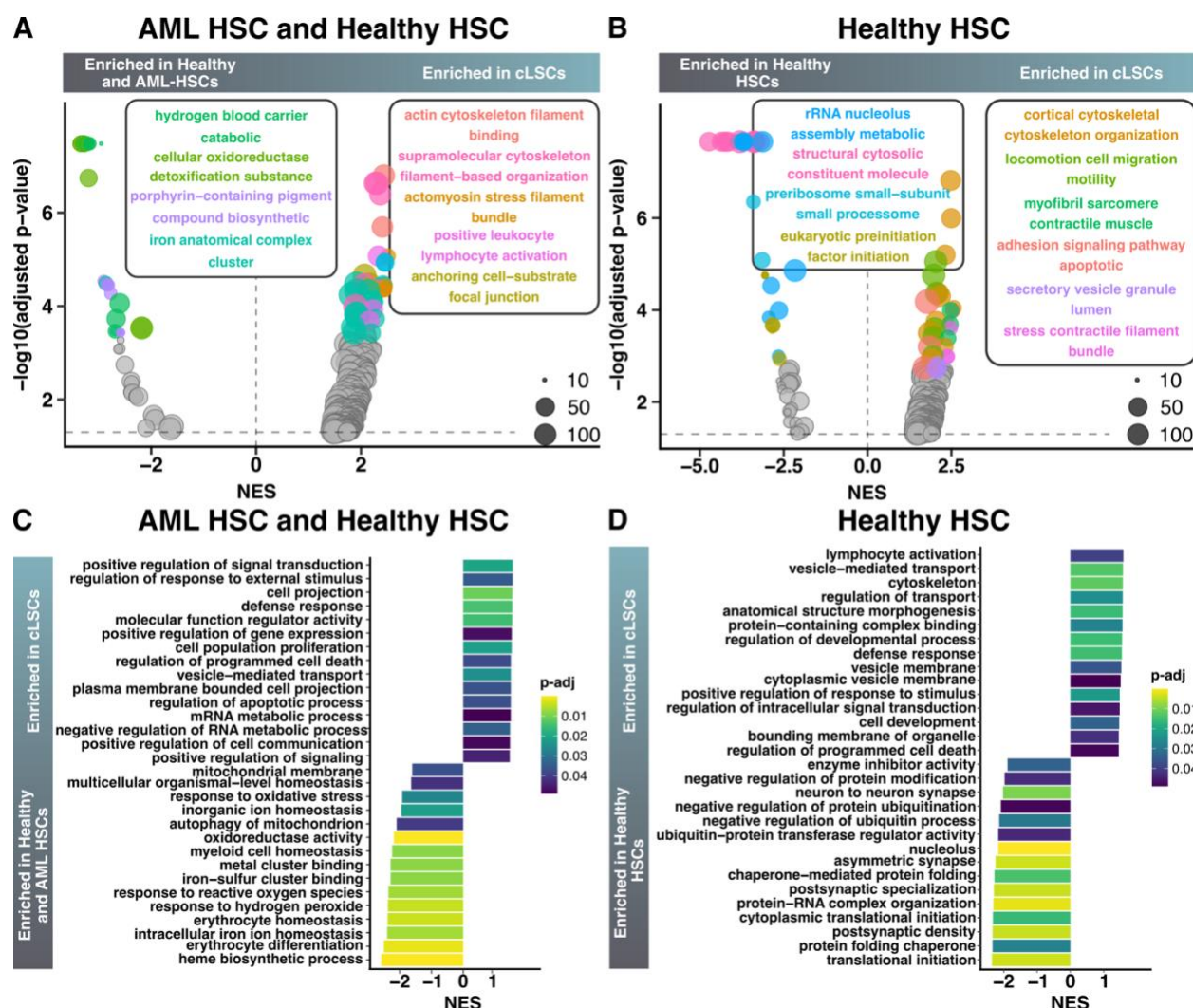


Figure S7: cLSCs have unique bimolecular properties A) Volcano plot showing enriched gene ontology terms based on GSEA between cLSC and AML and Healthy HSCs. Terms were clustered based on semantic similarity, and node color indicates cluster membership. Cluster labels represent the top terms based on frequent words present in the GO term description. Node size represents gene count. B) Volcano plot showing enriched gene ontology terms based on GSEA between cLSCs and Healthy HSCs alone. The plot is formatted similarly to panel A. C) Bar plot of enriched GO terms between cLSC and AML and Healthy HSCs. D) Bar plot of enriched GO terms between cLSCs and Healthy HSCs. Colors in panels C-D indicate adjusted p-value significance levels. GO – gene ontology, GSEA – gene set enrichment analysis.

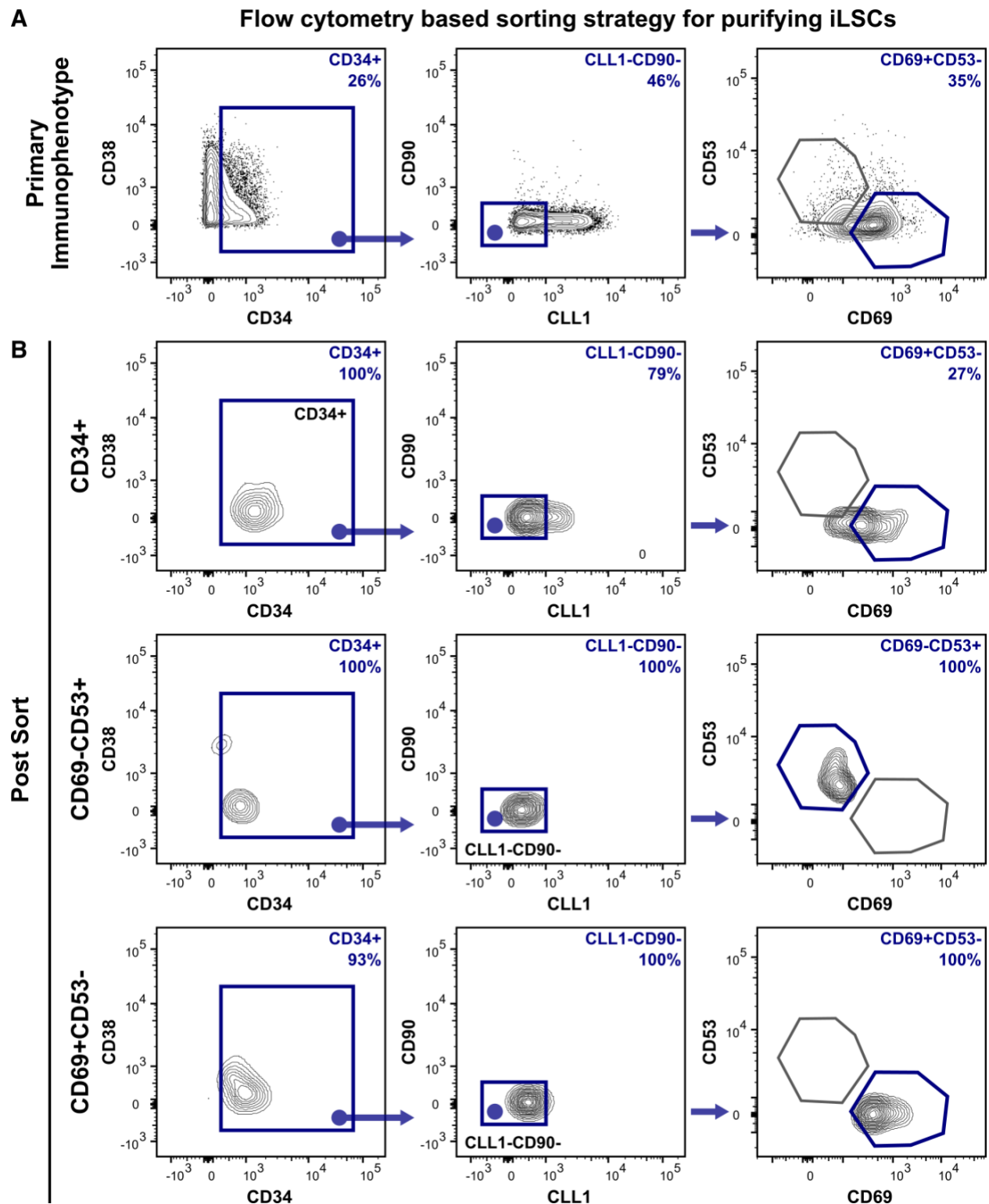
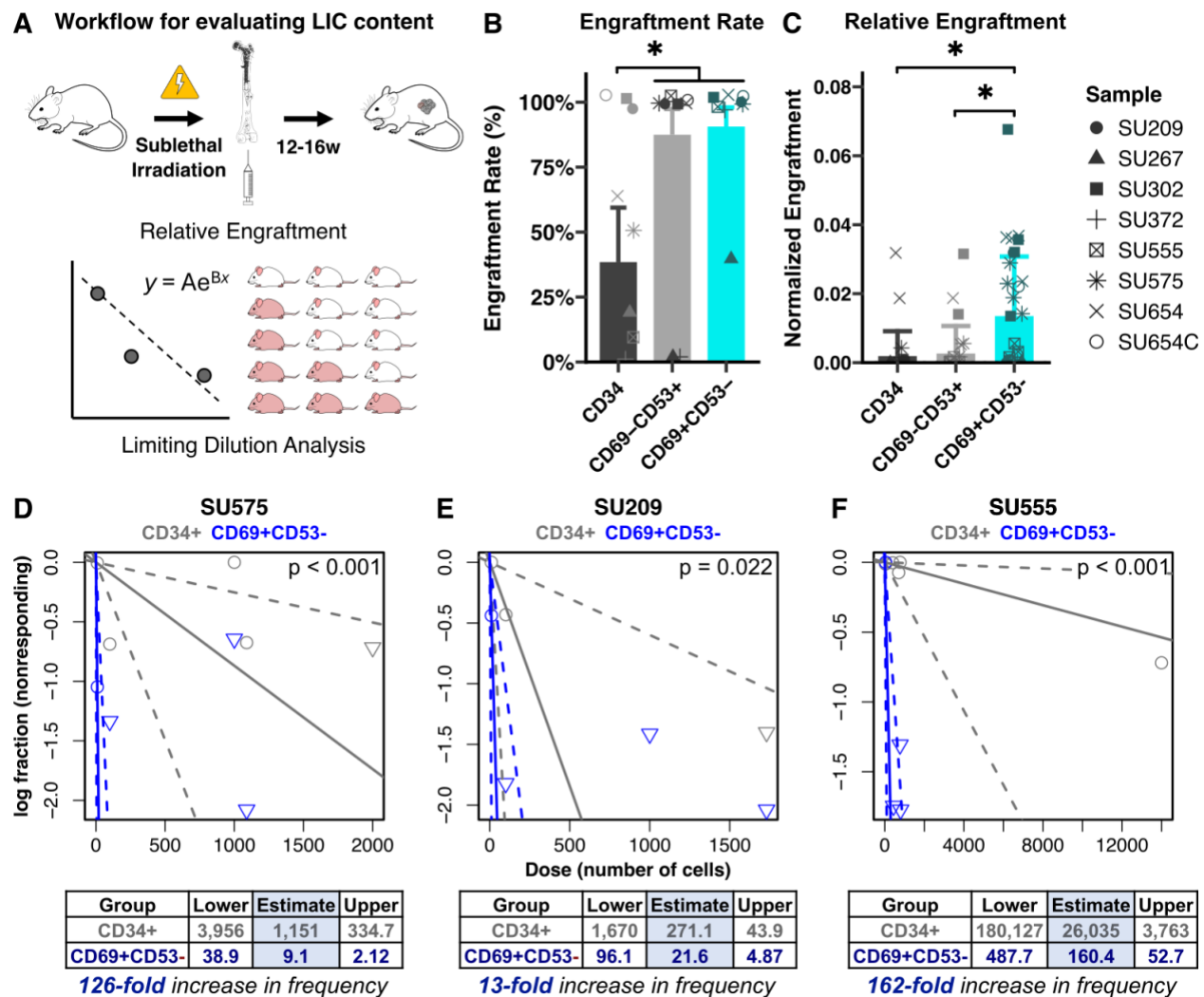


Figure S8: Representative plots of primary and post sort analysis of lymphocyte depleted primary AML cells. A) Sorting strategy for AML sub-populations including Lin-CD34+ AML, Lin-CD34+CLL1-CD69-CD53+, and Lin-CD34+CLL1-CD69+CD53- cells, with B) representative post sort purities for SU654.



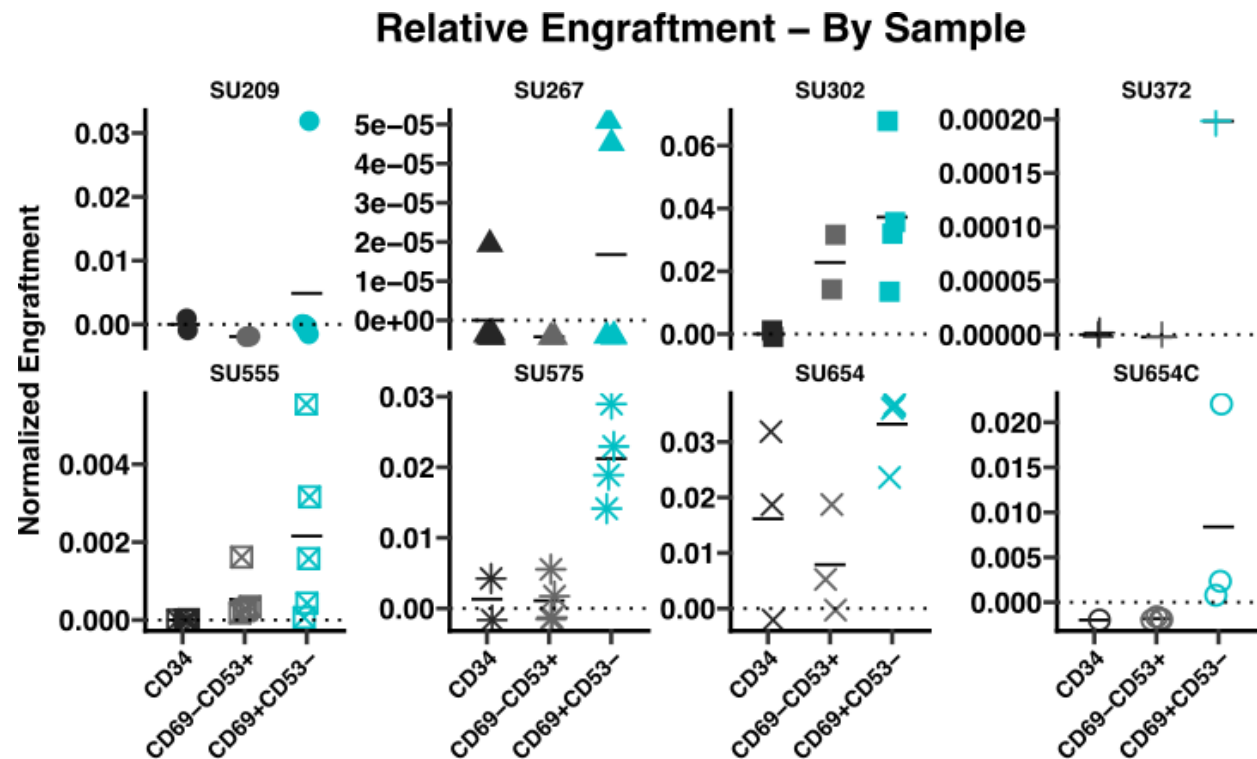


Figure S9: CD34+CLL1-CD90-CD69+CD53- iLSCs produce greater relative engraftment in primary transplants. The relative engraftment of purified subsets in the primary transplants for all 8 patients. Horizontal lines represent group means.

gene signature which was associated with worse outcomes independent of age, high and low risk mutations, cytogenetic risk categories, and validated prognostic AML signatures in a multi-variate analysis (Figure 1D). Importantly, hrLSC2 expression can further stratify patient survival within high or low risk mutation subgroups and cytogenetic risk categories (Figure 1F-H, Figure 3B-D). The strong performance of hrLSC2 highlights its potential utility as a biomarker for improving AML survival prediction and risk stratification. The added utility is likely due to orthogonal information captured by hrLSC2, which is highlighted further in our deconvolution analysis with CIBERSORTx (Figure 2-3). Using this approach, we were able to study hrLSC2 in a cell type specific context as we identified a population of AML cells, i.e. cLSCs and iLSCs, which express hrLSC2 at high levels (Figure 2), exhibit a unique immunophenotype (Figure S6H), and are highly prognostic in de novo AML (Figure 2-3). We also show that cLSCs are not only enriched in high-risk disease but also persist or expand at relapse (Figure 3E-J). Our analysis suggests that cLSCs are clinically high-risk and that aggressive treatment modalities including cytotoxic chemotherapy and HCT are ineffective at eliminating these cells. These observations are in line with the distinct yet complimentary role of the cancer stem cell model with the stochastic gene mutation model of leukemia pathogenesis⁶. Additional work will be required to prospectively validate this signature in patients receiving novel AML therapeutics.

The simplicity of a two gene signature allowed us to identify cLSCs based on co-expression of *HOPX* and *SOCS2* in scRNA-seq data (Figure 4J). Importantly, cells marked by high hrLSC2 expression differentially expressed surface markers compared to the bulk AML (Figure 4B), which allowed us to identify a sub-population of CD34+CLL1-CD90-CD69+CD53- iLSCs (Figure 4F). This immunophenotype was translated into a FACS-based sorting strategy (Figure S8) and used to purify iLSCs from primary AML samples. CD34+CLL1-CD90-CD69+CD53- iLSCs consistently produced greater engraftment in the xenotransplant compared to CD34+CLL1-CD90-CD69-CD53+ cells and bulk CD34+ cells (Figure 5C, Figure S9). The results from the LDA further support that CD34+CLL1-CD90-CD69+CD53- iLSCs contain greater LSC content (Figure 5D-F). In fact, we observed an LSC frequency upwards of 1/9 in this population, which is a significant improvement over previously published flow-based LSC sorting strategies in adult human AML. The association between hrLSC2 and LSC content is important, as the ability to prospectively isolate a highly pure LSC population across a diverse collection of AML cases has previously been an unmet goal. Indeed, while previously implicated LSC markers like CD99 were associated with cLSC content, markers like CLL1, TIM3, CD123, and CD25 had variable associations with hrLSC2 (Figure 2B). Of note, we show that CLL1+ cells significantly lose long term engraftment capacity in adult AML and healthy hematopoiesis⁴⁶, which argues against its role as a stem cell marker. The combination of hrLSC2 and the associated iLSC immunophenotype can serve as a valuable tool for identifying, tracking, and studying AML LSCs. For instance, hrLSC2 genes and surface markers can be translated into clinical assays for detecting measurable residual disease for AML patients^{5,59}, which can be evaluated in future prospective clinical trials.

Although we show that *HOPX* and *SOCS2* expression can mark LSCs, the underlying biological relevance of these genes remains unclear. *Hopx* is a Hox family member, and expression of *HOPX* is the strongest negative predictor of clinical outcomes in AML from prior meta-analyses^{8,60,61}. Prior work suggests that *Hopx* is both a marker of stem and progenitor cells in multiple organs⁶²⁻⁶⁴ and lymphocyte differentiation⁶⁵. *Hopx* has been historically difficult to study as it does not directly bind DNA but may interact as a complex with other transcription factors (TFs) and proteins⁶⁶. For instance, in cardiac development, *Hopx* has been shown to repress transcription through interactions with serum response factor and histone deacetylases (HDACs) in animal models⁶⁷. Follow up studies have shown that *Hopx* interacts with HDAC2 to impede cell cycle entry and proliferation⁶⁸. Chromatin immunoprecipitation experiments in murine hearts showed *Hopx* interactions in genomic regions proximal to Wnt family member genes, with *Hopx*

inhibition leading to repressed Wnt signaling⁶⁹. In murine hematopoiesis, *Hopx* disruption leads to reduced frequency of multipotent progenitors, decreased HSPC quiescence, and impaired engraftment⁷⁰. Studies in human pluripotent stem cells revealed that *HOPX* disruption leads to a block in primitive hematopoietic differentiation⁶². Molecular profiling in these studies revealed increased activity of Wnt target genes after *HOPX* disruption. Overall, it appears that *Hopx* modulates stem cell quiescence and function, in part through inhibiting Wnt signaling via interactions with HDAC/TF complexes.

The second gene, *SOCS2*, is a suppressor of cytokine signaling member and has been associated with worse outcomes in AML^{10,61,71,72}. *SOCS2* is upregulated upon activation of JAK/STAT and provides negative feedback in this signaling pathway. In the context of healthy hematopoiesis, the basal expression of *SOCS2* has been shown to be higher in HSCs compared to differentiated cells and is upregulated following cytokine stimulation⁷¹. In mice, *Socs2* deficiency was not only associated with unrestricted HSPC proliferation upon stress, but it also led to decreased HSC frequency and stem cell exhaustion over serial transplantation⁷¹. In myeloid malignancies, JAK/STAT signaling is often dysregulated due to the underlying inflammatory milieu and activation of STAT proteins^{73,74}. Additional work is needed to further characterize the role of both *SOCS2* and *HOPX* in adult human AML LSC pathogenesis.

iLSCs also express CD69, a type II transmembrane C-type lectin receptor, which has been implicated as a stem and progenitor cell marker in human hematopoiesis. We have shown that CD69+ multipotent progenitors (MPPs) have long term engraftment capacity and can be separated from erythroid and myeloid biased MPPs in healthy adult hematopoiesis⁴⁶. Prior work in *Mll-AF9/NRAS^{G12V}* murine AML revealed that CD36-CD69+ cells are enriched for LSC content⁷⁵, and follow up analysis in human AML samples showed that CD36-CD69+ cells had increased colony-forming capacity⁷⁶. Additional studies analyzing pediatric AML samples pre and post cytotoxic chemotherapy using scRNA-seq revealed a persistent CD69+ HSC-like subpopulation post treatment which also up-regulated *HOPX* and *SOCS2*⁷⁷. Our analysis supports the association of CD69 with LSC content in adult human AML using xenotransplantation assays which has not been shown previously. Like the hrLSC2 genes, the mechanistic association between CD69 and LSC pathogenesis is unclear. CD69 is known to be an activation marker of lymphocytes and platelets⁷⁸, but its role in stem cell biology is not understood. It has been shown to bind both carbohydrates and proteins, and can modulate cell metabolism, microenvironment interactions, and migration through both JAK/STAT and mTOR signaling. Although our GSEA analysis of cLSCs supports these overarching themes, future work will need to address the role of CD69 in stem cell biology.

In summary, our work provides three important advancements in the study of AML LSCs: 1) a parsimonious two-gene LSC signature, hrLSC2, which not only has a strong and independent association with poor outcomes in de novo AML but is also enriched in a specific AML subpopulation, 2) an LSC computational framework for studying AML, and 3) a new immunophenotype and sorting strategy which significantly improves the purification of LSCs in adult human AML.

Limitations of the study

This study provides significant advancements in our understanding of clinically relevant adult human AML LSCs. However, our analysis was limited by the scarcity of primary adult human AML tissue, as the frequency of LSCs in the majority of human AML is low, and the expense and effort of large-scale *in vivo* engraftment studies. Like prior studies, we had to functionally evaluate populations of cells, rather than single cells, for our xenotransplantation assays. This of course impairs our ability to definitively determine if our purified cells exist as a homogenous population

or as a collection of cells with various functional capabilities even though this approach is consistent with historical advancements and refinements in LSC immunophenotyping. Regardless of these limitations, we were able to observe significant differences amongst the purified cells which will be informative when designing and implementing translational research in human AML.

Resource Availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Ravindra Majeti (rmajeti@stanford.edu).

Materials availability

This study did not generate new unique reagents.

Data and Code Availability

- Single cell sequencing data have been deposited at GEO and are publicly available as of the date of publication. Accession numbers are listed in the key resources table.
- All original code used in this work is publicly available as of the date of the publication. The link is listed in the key resources table.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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Author contributions

Conceptualization, A.E., A.J.G. and R.M.; Methodology, A.E., A.J.G., A.M.N and R.M.; Investigation, A.E., Y.N., A.C.F., X.H., and B.B.; Writing – Original Draft, A.E. and R.M.; Writing – Review & Editing, A.E., A.J.G., A.M.N., and R.M.; Funding Acquisition, R.M.; Supervision, A.E., T.K., Y.K., D.K., M.H.L., A.J.G., A.M.N., and R.M.

Declaration of interests

R.M. is on the Advisory Boards of Kodikaz Therapeutic Solutions, Pheast Therapeutics, 858 Therapeutics, Prelude Therapeutics, Mubadala Capital, Aculeus Therapeutics, Sequentify, and

BMS. R.M. is a co-founder and equity holder of Pheast Therapeutics, and MyeloGene. A.M.N is co-founder of Cibermed Inc and A.J.G has consulted for them. All other authors declare no relevant competing interests.

Figure titles and legends

Key acronyms: AML – acute myeloid leukemia, cLSC – candidate leukemia stem cell, DC – dendritic cell, EMPP – erythroid-biased MPP, Ery – erythrocyte, GMP – granulocyte-monocyte progenitor, hrLSC2 – two gene high risk LSC signature, HSC – hematopoietic stem cell, HSPC – hematopoietic stem and progenitor cell, iLSC – immunophenotypic LSC, LMPP – lymphoid-primed MPP, LSC – leukemia stem cell, Mono – monocyte, MPP – multipotent progenitor, ProBEM – basophil eosinophil mast cell progenitor, ProB – B cell progenitor.

Figure 1: hrLSC2 is a prognostic gene signature in de novo AML. A) General workflow for identifying an LSC-specific gene signature in de novo AML. B) Key terminology used throughout the manuscript and Venn diagram illustrating relevant LSC sup-populations evaluated in this study. C) Results of iterative coxph regression analysis ($n=1000$) in de novo and normal karyotype AML evaluating the performance of hrLSC2 against previously published prognostic AML signatures for predicting survival. D) Multi-variate coxph survival analysis evaluating the association between prognostic AML signatures, age, high and low risk mutation status, and cytogenetic risk categories. E-H) Kaplan-Meier survival analysis evaluating hrLSC2 status in combination with established high and low risk features. Statistical significance was tested using either the Wald or Log-rank test.

Figure S1: Dataset curation, filtration, and analysis workflow for deriving hrLSC2.

Figure S2: Description of key dataset metrics. Principal component plot of gene expression labeled by (A) cohort and (B) assay subtype. C) Correlation between processed gene counts from RNA-seq and MA data with highly prognostic genes labeled. D) Bar plot showing relative proportions of each cohort in the training and test datasets. Comparison of training and test dataset compositions for key characteristics including (E) age, (F) survival, (G-J) clinical features, and (K) assay. MA – microarray, Neg – negative, Pos – positive, RNA – RNA-seq.

Figure S3: Single cell RNA-seq dataset curation, processing, annotation, and integration workflow.

Figure 2: cLSCs are the most prognostic characterized cell type in de novo AML. A) Workflow for constructing an integrated scRNA-seq dataset for de novo AML. B) Post integration UMAP projection of cells from AML patients and healthy donors annotated using an HSPC-specific classifier. C) hrLSC2 expression in AML cells based on normal cell type annotations. D) Expression of *HOPX* and *SOCS2* for AML cells annotated as HSCs. Percentage reflects cells that co-express *HOPX* and *SOCS2*. E) Histogram illustrating hrLSC2 expression for AML cells annotated as HSCs. Dashed red line indicates threshold for annotating cLSCs. F-G) Expression of *HOPX* and *SOCS2* for cells

re-annotated based on hrLSC2 expression. H) UMAP projection of the final integration after updating cell type annotations for hrLSC2 expression. I) Results of iterative Cox regression analysis ($n=1000$) in de novo AML evaluating the performance of cell type content for predicting survival. J) Kaplan-Meier survival analysis evaluating cLSC content. Statistical significance was tested using either the Wald or Log-rank test.

Figure S4: Constructing a high-resolution single cell reference for bulk gene expression deconvolution. A) Workflow for constructing a high-resolution signature matrix for a cell type specific survival analysis in de novo AML. B) Histogram of hrLSC2 expression across all AML cells. Dashed blue line indicates threshold for identifying cells expressing high levels of hrLSC2. C) Cell type frequencies within all cells and the hrLSC2 high (H) sub-population. Comparison of (D) Shannon diversity indices and (E) spread by sample for the hrLSC2 high sub-population and all cell types. The comparison containing all cells was down sampled to ensure the same number of cells. Each dot represents a single donor and is colored based on median hrLSC2 expression and sized based on cell number. F) Loss comparison over training epochs for both the original and final integration with scGen. G) Heatmap illustrating imputation results from deconvolving artificial bulk transcriptomes derived only from cells in each respective cluster. H) Comparison between imputation results and true cell type proportions from deconvolving artificial bulk transcriptomes containing mixtures of cell types at known frequencies. I) Deconvolution results of 942 de novo AML cases using CIBERSORTx. The violin plot presents cell type frequencies, and the heatmap is colored based on cohort (rows) and disease (columns).

Figure 3: cLSCs are associated with worse outcomes in de novo AML and expand in relapsed disease. A) Multi-variate coxph survival analysis evaluating the association between cLSC content, prognostic AML signatures, age, high and low risk mutation status, and cytogenetic risk categories. B-D) Kaplan-Meier survival analysis evaluating cLSC content in combination with established high and low risk features. cLSC content at diagnosis and relapse after (E) chemotherapy and (F) HCT. cLSC content for patients receiving chemotherapy or HCT at (G) diagnosis and (H) relapse. Red line highlights median values. P-values and effect sizes (Cohen's d) are indicated. I) Mean cell type proportions at diagnosis and relapse for patients receiving either chemotherapy or HCT. J) Mean proportion of top prognostic cell populations at diagnosis and relapse for patients receiving either chemotherapy or HCT. Statistical significance was tested using either the Wald or Log-rank test for panels A-D. Statistical significance was tested using t-test with multiple hypothesis correction for panels E-J. ***Adjust p-value < 0.0001, **Adjusted p-value < 0.001, *adjusted p-value < 0.05. Chemo – chemotherapy, Dx – diagnosis, HCT – hematopoietic cell transplant, Rel – relapse.

Figure 4: Cells expressing high hrLSC2 also express a unique immunophenotype. A) Schematic illustrating the workflow for identifying an immunophenotype for cLSCs (i.e. iLSCs) using a targeted gene expression panel and scADT-seq. B) Plot illustrating ADT rank and mean coefficient from Lasso regressions by sample comparing ADT expression and hrLSC2 status. C-E) Computational sorting strategy for purifying iLSCs. Normalized expression of F) hrLSC2, G) LSC17, and (H) LinClass7 within each computationally purified sub-population. I) Diversity measured by the Shannon diversity index for each purified sub-population. All comparisons are statistically significant unless marked by a

solid line. UMAP projections illustrating (J) hrLSC2 expression or sub-population gating based on the (K) updated sort or (L) prior gating strategy. Statistical significance was tested using a Wilcoxon ranked sum test with multiple comparison adjustments with an adjusted p-value < 0.05 considered significant. scADT-seq – single cell antibody directed tag sequencing, G1 – gate 1, G2 – gate 2.

Figure S5: hrLSC2 expression can be used to mark a sub-population of AML cells. Cellular expression of (A) hrLSC2, (B) LSC17, and (C) LinClass7 for each donor. D) Scatter plot comparing *HOPX* and *SOCS2* expression, where each cell is colored based on its calculated hrLSC2 score. E) Number of non-zero coefficients using Lasso regression comparing ADTs and hrLSC2 as a continuous variable (Con) or using Lasso classification based on quantile thresholding for determining cLSC status. A threshold of 75% (i.e. the top 25% being considered cLSCs) was determined to be optimal as there is a significant decrease in coefficients with higher thresholding. F) Scatter plot comparing *HOPX* and *SOCS2* expression and cLSC status based on top 25% thresholding. G-I) Expression of gene signatures based on cLSC status. Statistical significance was tested using a t test or ANOVA with multiple comparison adjustments. *Adjusted p < 0.05.

Figure S6: Computationally purifying iLSCs can increase sub-population hrLSC2 expression and purity. Results from purifying AML cells using whole transcriptome scRNA-seq and scADT-seq. Normalized expression of (A) hrLSC2, (B) LSC17, and (C) LinClass7 within each purified sub-population. D) Diversity measured by the Shannon diversity index for each purified sub-population. UMAP projections illustrating (E) hrLSC2 expression or sub-population gating based on the (F) updated computational defined sub-populations or (G) prior LSC gating strategies. H) *CD53* and *CD69* gene expression for cells computationally purified from the scRNA-seq cohort (Figure 2) based on negative lineage marker (Lin-), *CD34*, and *CLEC12A* (gene encoding for CLL1) expression which corresponds to G2. G1 – gate 1, G2 – gate2.

Figure S7: cLSCs have unique bimolecular properties A) Volcano plot showing enriched gene ontology terms based on GSEA between cLSC and AML and Healthy HSCs. Terms were clustered based on semantic similarity, and node color indicates cluster membership. Cluster labels represent the top terms based on frequent words present in the GO term description. Node size represents gene count. B) Volcano plot showing enriched gene ontology terms based on GSEA between cLSCs and Healthy HSCs alone. The plot is formatted similarly to panel A. C) Bar plot of enriched GO terms between cLSC and AML and Healthy HSCs. D) Bar plot of enriched GO terms between cLSCs and Healthy HSCs. Colors in panels C-D indicate adjusted p-value significance levels. GO – gene ontology, GSEA – gene set enrichment analysis.

Figure S8: Representative plots of primary and post sort analysis of lymphocyte depleted primary AML cells. A) Sorting strategy for AML sub-populations including Lin-CD34+ AML, Lin-CD34+CLL1-CD69-CD53+, and Lin-CD34+CLL1-CD69+CD53- cells, with B) representative post sort purities for SU654.

Figure 5: CD34+CLL1-CD90-CD69+CD53- iLSCs are enriched for LSC content. A) Schematic illustrating the workflow for xenotransplantation assays. B) Engraftment rate

and C) relative engraftment between purified subsets in the primary transplants from 8 patient samples. D-F) Results from the limiting dilution analysis for 3 AML cases and significance was determined using the chi-squared test. Statistical significance for the other comparisons was tested using ANOVA with multiple comparison adjustments. *Adjusted p-value < 0.05.

Figure S9: CD34+CLL1-CD90-CD69+CD53- iLSCs produce greater relative engraftment in primary transplants. The relative engraftment of purified subsets in the primary transplants for all 8 patients. Horizontal lines represent group means.

STAR Methods

KEY RESOURCE TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
CD3	BD Biosciences	Cat# 555341, RRID: AB_395747
CD8	BD Biosciences	Cat# 555368, RRID: AB_395771
CD19	BD Biosciences	Cat# 555414, RRID: AB_395814
CD20	BD Biosciences	Cat# 555624, RRID: AB_395990
GPA (CD235a)	BD Biosciences	Cat# 559944, RRID: AB_397387
CD34	BD Biosciences	Cat# 340667, RRID: AB_400531
CD90	BD Biosciences	Cat# 555596, RRID: AB_395970
CD69	BD Biosciences	Cat# 562884, RRID: AB_2687422
CLL1	BD Biosciences	Cat# 562569, RRID: AB_11154053
CD53	BD Biosciences	Cat# 745753, RRID: AB_2743219
CD45	BD Biosciences	Cat# 560367, RRID: AB_1645573
Anti-mouse CD45.1	eBioscience	Cat# 11-0453-85, RRID: AB_465059
Anti-mouse TER-119	eBioscience	Cat# 47-5921-82, RRID: AB_1548786
CD33	BD Biosciences	Cat# 555450, RRID: AB_395843
HLA-ABC	BD Biosciences	Cat# 568004, RRID: AB_2916810
CD10 AbSeq Oligo AHS0051	BD Biosciences	Cat# 940045, RRID: AB_2875936
CD117 AbSeq Oligo AHS0064	BD Biosciences	Cat# 940051, RRID: AB_2875942
CD11b AbSeq Oligo AHS0005	BD Biosciences	Cat# 940008, RRID: AB_2875899
CD123 AbSeq Oligo AHS0020	BD Biosciences	Cat# 940020, RRID: AB_2875911
CD14 AbSeq Oligo AHS0037	BD Biosciences	Cat# 940005, RRID: AB_2875896
CD16 AbSeq Oligo AHS0053	BD Biosciences	Cat# 940006, RRID: AB_2875897
CD184 AbSeq Oligo AHS0060	BD Biosciences	Cat# 940056, RRID: AB_2875947
CD19 AbSeq Oligo AHS0030	BD Biosciences	Cat# 940004, RRID: AB_2875895

CD196 AbSeq Oligo AHS0034	BD Biosciences	Cat# 940033, RRID: AB_2875924
CD2 AbSeq Oligo AHS0029	BD Biosciences	Cat# 940046, RRID: AB_2875937
CD20 AbSeq Oligo AHS0008	BD Biosciences	Cat# 940016, RRID: AB_2875907
CD235a_b AbSeq Oligo AHS0048	BD Biosciences	Cat# 940040, RRID: AB_2875931
CD24 AbSeq Oligo AHS0042	BD Biosciences	Cat# 940028, RRID: AB_2875919
CD244 AbSeq Oligo AHS0131	BD Biosciences	Cat# 940362, RRID: AB_2876232
CD25 AbSeq Oligo AHS0026	BD Biosciences	Cat# 940009, RRID: AB_2875900
CD3 AbSeq Oligo AHS0033	BD Biosciences	Cat# 940000, RRID: AB_2875891
CD32 AbSeq Oligo AHS0073	BD Biosciences	Cat# 940069, RRID: AB_2875960
CD33 AbSeq Oligo AHS0044	BD Biosciences	Cat# 940031, RRID: AB_2875922
CD34 AbSeq Oligo AHS0061	BD Biosciences	Cat# 940021, RRID: AB_2875912
CD38 AbSeq Oligo AHS0022	BD Biosciences	Cat# 940013, RRID: AB_2875904
CD4 AbSeq Oligo AHS0032	BD Biosciences	Cat# 940001, RRID: AB_2875892
CD44 AbSeq Oligo AHS0054	BD Biosciences	Cat# 940039, RRID: AB_2875930
CD45 AbSeq Oligo AHS0040	BD Biosciences	Cat# 940002, RRID: AB_2875893
CD45RA AbSeq Oligo AHS0009	BD Biosciences	Cat# 940011, RRID: AB_2875902
CD56 AbSeq Oligo AHS0019	BD Biosciences	Cat# 940007, RRID: AB_2875898
CD69 AbSeq Oligo AHS0010	BD Biosciences	Cat# 940019, RRID: AB_2875910
CD7 AbSeq Oligo AHS0043	BD Biosciences	Cat# 940029, RRID: AB_2875920
CD8 AbSeq Oligo AHS0027	BD Biosciences	Cat# 940003, RRID: AB_2875894
CD86 AbSeq Oligo AHS0057	BD Biosciences	Cat# 940025, RRID: AB_2875916
CD9 AbSeq Oligo AHS0082	BD Biosciences	Cat# 940078, RRID: AB_2875969
CD90 AbSeq Oligo AHS0045	BD Biosciences	Cat# 940032, RRID: AB_2875923
CD93 AbSeq Oligo AHS0124	BD Biosciences	Cat# 940215, RRID: AB_2876097
CD99 AbSeq Oligo AHS0123	BD Biosciences	Cat# 940214, RRID: AB_2876096
CLL1 AbSeq Oligo AHS0097	BD Biosciences	Cat# 940212, RRID: AB_2876094
Tim3 AbSeq Oligo AHS0016	BD Biosciences	Cat# 940066, RRID: AB_2875957
CD36 AbSeq Oligo AHS0135	BD Biosciences	Cat# 940224, RRID: AB_2876105
CD52 AbSeq Oligo AHS0463	BD Biosciences	Cat# 460435, RRID: Custom
CD53 AbSeq Oligo AHS0326	BD Biosciences	Cat# 460418, RRID: Custom
CD74 AbSeq Oligo AHS0325	BD Biosciences	Cat# 460416, RRID: Custom
Critical commercial assays		
BD Rhapsody Cartridge reagent kit	BD Biosciences	Cat# 633731
Deposited data		
Human AML BMMC scRNA-seq and scADT-seq	This paper	GEO: GSE313060

Software and algorithms		
Seurat R package (v3.5.1 and v4.4.0)	Butler et al. (2018); Stuart et al. (2019)	https://cran.r-project.org/web/packages/Seurat/index.html
CIBERSORTx (v1.0)	Newman et al. (2019)	https://cibersortx.stanford.edu/
BD Rhapsody™ WTA Analysis Pipeline (v1.2)		www.sevenbridges.com
STAR (v2.7.10b)	Dobin et al. (2013)	https://github.com/alexdobin/STAR
RSEM (v1.3.3)	Li and Dewey (2011)	https://github.com/deweylab/RSEM
NMF R package (v0.20.0)	Gaujoux and Seoighe (2010)	https://cran.r-project.org/web/packages/NMF/index.html
factoextra R package (v1.0.7)	Kassambara et al 2020	https://cran.r-project.org/web/packages/factoextra/index.html
affy R package (v1.86.0)	Gautier et al 2004	https://bioconductor.org/packages/release/bioc/html/affy.html
ape R package (v5.4.1)	Paradis and Schliep (2019)	https://cran.r-project.org/web/packages/ape/index.html
caret R package (v6.0)	Kuhn et al 2023	https://cran.r-project.org/web/packages/caret/index.html
glmnet R package (v4.1.9)	Friedman et al 2010	https://cran.r-project.org/web/packages/glmnet/index.html
Random ForestSRC (v3.4.3)	Ishwaran and Kogalur 2021	https://github.com/kogalur/randomForestSRC/
DoubletFinder R package (v2.0)	McGinnis et al 2019	https://github.com/chris-mcginis-ucsf/DoubletFinder
SoupX R package (v1.6.2)	Young et al 2020	https://cran.r-project.org/web/packages/SoupX/index.html
mixtools R package (v2.0.0)	Young et al 2022	https://cran.r-project.org/web/packages/mixtools/index.html
vegan R package (v2.6.4)	Oksanen et al 2022	https://cran.r-project.org/web/packages/vegan/index.html
clusterProfiler (v4.10.0)	Wu et al 2021	https://guangchuangyu.github.io/software/clusterProfiler/
TopKLists (v1.0.8)	Schimek et al 2022	https://cran.r-project.org/web/packages/TopKLists/index.html
R (v3.6.2)	R CRAN	https://cran.r-project.org/
Prism (v7)	GraphPad Software, Inc.	N/A

FlowJo		
Other		
Resource website for this work	This paper	https://github.com/systemsheme/hrisc2
BEAT AML RNA-seq	Tyner et al 2018	dbGaP: phs001657
AML RNA-seq	Jayavelu et al 2022	EGA: EGAD00001008484
AML RNA-seq	McKeown et al	SRA: PRJNA381888
TCGA-LAML RNA-seq	Ley et al 2013	dbGAP: phs000178
AML RNA-seq	Hackl et al 2015	GSE: GSE66525
AML RNA-seq	Christopher et al 2018	dbGaP: phs000159.v11.p5
AML RNA-seq	Li et al 2016	dbGaP: phs001027.v1.p1
AML RNA-seq	Toffalori et al 2019	EBI: E-MTAB-7456
Healthy RNA-seq	Corces et al 2016	GSE: GSE74246
AML Microarray	Metzeler et al 2008	GSE: GSE12417
AML Microarray	Tomasson et al 2008	GSE: GSE10358
AML Microarray	Wouters et al 2009	GSE: GSE14468
AML scRNA-seq	Abbas et al 2021	EGA: EGAD00001007672
AML scRNA-seq	Beneyto-Calabuig et al 2023	EGA: EGAS00001007078
AML scRNA-seq	Bordeleau et al 2024	GSE: GSE241989
AML scRNA-seq	Ennis et al 2023	GSE: GSE227903
AML scRNA-seq	Ianevski et al 2021	EGA: EGAS00001004444
AML scRNA-seq	Lasry et al 2023	GSE: GSE185381
AML scRNA-seq	Pei et al 2023	GSE: GSE232559
AML scRNA-seq	Petti et al 2019	dbGaP: phs000159.v11.p5
AML scRNA-seq	Stiff et al 2024	GSE: GSE266498
AML scRNA-seq	Van Galen et al 2019	GSE: GSE116256
AML scRNA-seq	Zheng et al 2017	https://www.10xgenomics.com/dataset

METHOD DETAILS

Description of healthy adult donors

Bone marrow mononuclear cells were obtained from patients with AML with informed consent and compliance with relevant ethical regulations (Stanford University IRB 6453). All cells used in this study were cryopreserved and subsequently thawed by the dropwise addition of IMDM (GIBCO, cat. no. 12440-061) containing 20% fetal bovine serum (FBS; Sigma, cat. no. F1051).

Curation of bulk gene expression data in adult human AML

Microarray and RNA-seq data were normalized to unify data from diverse platforms using established methods⁶¹. For RNA-seq data, raw fastq or bam files were downloaded and aligned to the human reference genome (GRCh38) using STAR (v2.7.10b)⁷⁹ and count tables with transcripts per million (TPM) normalization were generated using RSEM (v1.3.3)⁸⁰. For microarray data, CEL files were downloaded and processed using the Robust Multi-array Average algorithm with the affy R package (v1.86.0)⁸¹. Gene counts were converted to z-values and scaled by study to facilitate cross study and platform comparisons.

Derivation of hrLSC2

AML patients with gene expression data were filtered for those that with survival > 30 days to minimize confounding due to initial disease severity (Table ST1). We also included only de novo AML at diagnosis (excluding acute promyelocytic leukemia). Data were then split into a training and test set (2:1 split), and subset for LSC genes present in all datasets (Table ST2). Iterative univariate coxph regressions (n=1000) were performed comparing gene expression and survival. We also performed Lasso regression using the glmnet R package⁸², random forest (RF) using the randomForestSRC R package⁸³, and recursive feature elimination using RF with the caret R package⁸⁴ to identify top features. High performing features were then evaluated using ranked list analysis using TopKLists⁴². The top two features, *HOPX* and *SOCS2*, were used to create the final model and coefficients were estimated using coxph ($0.4 \cdot HOPX + 0.2 \cdot SOCS2$). This model was then compared to established AML gene signatures and standard clinical variables using coxph in both univariate and multi-variate models.

Single cell captures for whole transcriptome and surface marker quantification

Cryopreserved adult bone marrow cells were thawed and stained with annexin V (Thermo Fisher Scientific) per protocol for 15 min. Cells were resuspended with DAPI and annexin V/DAPI negative cells were sorted for subsequent ADT staining and single cell capture per established protocols (BD Rhapsody™). The BD Rhapsody™ workflow is a nanowell based single cell capture system with established protocols for generating single cell whole transcription and ADT cDNA libraries. Briefly, an ADT cocktail (1uL per ADT) was prepared in a total of 100 uL of buffer. Cells were first washed and then labeled with the ADT cocktail for 30 min at 4C, washed 3 times, resuspended in sample buffer (BD Rhapsody™ Cartridge reagent kit), and captured with the BD Rhapsody™ multiomic single cell system using the manufacturer's instructions⁸⁵. ADT and whole transcriptome libraries were generated per manufacture's protocol, evaluated by qubit and bioanalyzer, pooled and sequenced using NextSeq500 or Illumina Novaseq S2 (Illumina). Sequencing depth was determined based on manufacturer's recommendations.

Sample processing, integration, and quality control

We applied stringent quality control that incorporates the basic Seurat workflow for multimodal data processing and integration^{86,87}, ambient RNA and ADT removal^{88,89}, doublet discrimination using DoubletFinder⁹⁰, and elimination of cells expressing mutually exclusive lineage markers based on prior work⁴⁶. Fastq files generated from the BD Rhapsody™ workflow were processed on Seven Bridges (<https://www.sevenbridges.com>) using the standard analysis pipeline (BD Rhapsody™ WTA Analysis Pipeline) and a custom ADT fasta (Table ST10). The mRNA panel for the surface immunophenotype analysis (Figure 4, S5) was derived from differentially expressed genes from FACS sorted HSPC sub-populations, LSCs, Blasts, and normal effector cells⁵⁷. The top ranked genes associated with these cell types were consolidated into a 500 gene panel. The ADT panel against common surface epitopes associated with healthy hematopoietic and leukemia subsets (Table ST10). The unique molecular identifier (UMI) count matrices were imported into R (v3.6.2) and processed using Seurat (v3.5.1 and v4.4.0)⁸⁷. Each domain, ie mRNA and ADT counts, was normalized using log-normalization or centered using log ratio normalization respectively for visualization and comparative analyses. A third concatenated dataset was created by merging normalized counts for specific down-stream analyses. Poor quality cells were removed based on total RNA content and percent mitochondrial content using the top 1% expression level. Doublet discrimination was first performed using DoubletFinder with standard metrics⁹⁰. Doublets were removed and gene and surface marker expression were corrected for ambient noise using SoupX⁸⁸. DoubletFinder was used to filter doublets rather than a maximum RNA content filter due to the inherent variability of RNA content and celltype⁹¹. Mutually exclusive lineage markers (Table ST6) were subsequently used to remove heterotypic doublets, and poor-quality cells resulting from non-specific ADT binding^{46,92}. The filtered cells were subsequently

integrated by sample using canonical correlation analysis (CCA) and the resulting integrated matrix was used for downstream computational analyses. For the large-scale meta-analysis, scRNA-seq count data was processed using a similar workflow (Figure S3, Table ST7). Poor quality cells were removed using sample specific mitochondrial content analysis which was generally based on the 95th percentile as different single cell capture methods preferentially increase mitochondrial content. Each dataset was then labeled using an HSPC specific classifier⁴⁶ and integrated using scGen with standard parameters and the HSPC annotation⁴⁷.

Single cell hrLSC2 quantification and immunophenotype derivation

Per cell hrLSC2 expression was quantified using the same coefficients described above. Cells that ranked above the top 25% expression were labeled as cLSCs and used to determine associated features based on lasso regressions. The 25% threshold was determined by performing lasso regressions using a range of thresholds. The ability to identify informative coefficients dropped significantly at thresholds higher than 25% (Figure S5). Markers were ranked based on importance derived from the lasso analysis and evaluated manually. Candidate features were subsequently evaluated using FACS to identify markers with sufficient dynamic range to efficiently purify each sub-population.

Clustering and diversity analysis

Cells were computationally purified using the specific sorting strategies based on ADT counts. Gating thresholds were determined by modeling the expression of each ADT using mixture models ($k=2-3$) and the expectation maximization algorithm with mixtools⁴⁸. Sub-population purity was assessed using the Shannon diversity index and the vegan R package⁹³. Cellular spread was determined based on the mean Euclidean distance for each cell to the sample centroid using the principal component space.

Deconvolution of bulk AML gene expression data using CIBERSORTx

Each AML cohort was deconvolved using an hierarchal strategy⁹⁴ by serially applying two signature matrices. The first signature matrix was generated using bulk gene expression data from purified hematopoietic populations⁵⁷, and was used to differentiate T, B and NK cells from HSPC, myeloid, and erythroid subsets. The second signature matrix was generated using the single cell RNA-seq data (Figure 2A) by first converting single-cell expression values to transcripts per million (TPM) space. Cells were then split into a training and test dataset using a 2:1 split by cell type. The training set was used to generate a signature matrix using CIBERSORTx using the recommended parameters⁴⁹. Deconvolution performance was evaluated by building artificial bulk transcriptomes using the test dataset. We created pure cell type transcriptomes by combining cells based on their cell annotation, and artificial mixtures by combining cells based on randomly generated cell type frequencies based on their annotation ($n=100$). These mixtures contain cell types based on known frequencies, i.e. 1 for the pure transcriptomes or a known fraction based on the mixture. The counts from the pooled cells were converted to TPM and summed across all cells. The performance of the signature matrix was evaluated by comparing the imputed frequencies after deconvolution to the ground truth. No batch correction yielded the best results and was used for subsequent deconvolution of the bulk AML transcriptomes. The final cell type fractions were determined by multiplying the non-lymphocyte populations with the cell type content imputed by the second reference matrix.

Gene set enrichment analysis (GSEA)

GSEA was performed by differentially expressed genes using the integrated data and the FindMarkers function and the MAST test in Seurat. Markers that were differentially expressed at an adjusted p-value of less than 0.05 were evaluated by GSEA using GO terms with clusterProfiler

in R⁹⁵. Significant terms were clustered based on semantic similarity using the treeplot function for visualization purposes (Figure 5).

Animal care

All mouse experiments were conducted in accordance with a protocol approved by the Institutional Animal Care and Use Committee (Stanford Administrative Panel on Laboratory Animal Care #22264) and in adherence with the U.S. National Institutes of Health's Guide for the Care and Use of Laboratory Animals.

Fluorescence-activated cell sorting

Thawed cells were washed with FACS buffer (PBS, 2% FBS, 2 mM EDTA) and stained with antibodies for 30 minutes on ice in 50 uL total volume. Cells were then stained with Zombie Red for viability assessment for 5 minutes and subsequently washed prior to analysis. All cell sorting steps were confirmed using post-sort purity analyses. Flow cytometry was performed on a FACS Aria II (Becton Dickinson). Gates were drawn by internal positive and negative controls using healthy cells and validated by back-gating on marker positive events. For CD69, gating was further evaluated using fluorescence minus one gating. Gates drawn for all other markers were consistent across all samples (Figure S7) and the panels used are listed in Table ST9.

Xenotransplantation assays

Purified cells were transplanted into the right femur of sub-lethally irradiated NOD/SCID/IL2Rγ^{-/-}3/GM/SF (NSGS) mice (6-8 weeks, male and female, 200 rad 2-24 hours pre-transplant)⁵⁸. Cell dosing is provided in Table ST9. Mice were harvested at 10-18 weeks by crushing the pelvis and bilateral femurs and tibias. Cells were filtered and stained using the engraftment panel (Table ST9) to assess both engraftment and lineage. Human to mouse chimerism was calculated based on hCD45+HLA-ABC+CD33+CD19- cells relative to mouse CD45+mTer119- cells. Chimerism >0.1% was considered positive for long term engraftment. For comparison across samples (Figure 5F-G), chimerism was normalized to the sum of cell dose and time (weeks). Limiting dilution analysis was performed using cell dosing presented in Table ST9 and quantified using Extreme Limiting Dilution Analysis software⁹⁶.

Quantification and Statistical Analysis

Details of statistical tests performed can be found in figure legends or materials and methods. For significance testing, a t-test (2-tailed distribution and two-sample equal variance) with Mann-Whitney multiple comparison adjustment was used for comparing 2 conditions or a one-way ANOVA with Tukey's multiple comparison adjustment when comparing >2 conditions unless otherwise stated. Results with adjusted p-values <0.05 were considered significant. All statistical analyses were performed in R or Graphpad Prism.

References:

1. Surveillance, E., and End Results (SEER) Program (www.seer.cancer.gov) SEER*Stat Database: Mortality - All COD, Aggregated With State, Total U.S. (1969-2022). Cancer Stat Facts: Leukemia - Acute Myeloid Leukemia. 2024 ed.
2. Bataller, A., Kantarjian, H., Bazinet, A., Kadia, T., Daver, N., DiNardo, C.D., Borthakur, G., Loghavi, S., Patel, K., and Tang, G. (2024). Outcomes and genetic dynamics of acute myeloid leukemia at first relapse. *Haematologica* 109, 3543.
3. Döhner, H., Estey, E., Grimwade, D., Amadori, S., Appelbaum, F.R., Büchner, T., Dombret, H., Ebert, B.L., Fenaux, P., Larson, R.A., et al. (2017). Diagnosis and management of AML

- in adults: 2017 ELN recommendations from an international expert panel. *Blood* **129**, 424-447. 10.1182/blood-2016-08-733196.
4. Döhner, H., Wei, A.H., Appelbaum, F.R., Craddock, C., DiNardo, C.D., Dombret, H., Ebert, B.L., Fenaux, P., Godley, L.A., Hasserjian, R.P., et al. (2022). Diagnosis and management of AML in adults: 2022 recommendations from an international expert panel on behalf of the ELN. *Blood* **140**, 1345-1377. 10.1182/blood.2022016867.
5. Ediriwickrema, A., Gentles, A.J., and Majeti, R. (2023). Single-cell genomics in AML: extending the frontiers of AML research. *Blood* **141**, 345-355. 10.1182/blood.2021014670.
6. Kreso, A., and Dick, J.E. (2014). Evolution of the cancer stem cell model. *Cell stem cell* **14**, 275-291.
7. Thomas, D., and Majeti, R. (2017). Biology and relevance of human acute myeloid leukemia stem cells. *Blood* **129**, 1577-1585. 10.1182/blood-2016-10-696054.
8. Gentles, A.J., Plevritis, S.K., Majeti, R., and Alizadeh, A.A. (2010). Association of a leukemic stem cell gene expression signature with clinical outcomes in acute myeloid leukemia. *Jama* **304**, 2706-2715. 10.1001/jama.2010.1862.
9. Eppert, K., Takenaka, K., Lechman, E.R., Waldron, L., Nilsson, B., van Galen, P., Metzeler, K.H., Poepl, A., Ling, V., Beyene, J., et al. (2011). Stem cell gene expression programs influence clinical outcome in human leukemia. *Nature medicine* **17**, 1086-1093. 10.1038/nm.2415.
10. Ng, S.W., Mitchell, A., Kennedy, J.A., Chen, W.C., McLeod, J., Ibrahimova, N., Arruda, A., Popescu, A., Gupta, V., Schimmer, A.D., et al. (2016). A 17-gene stemness score for rapid determination of risk in acute leukaemia. *Nature* **540**, 433-437. 10.1038/nature20598.
11. Bonnet, D., and Dick, J.E. (1997). Human acute myeloid leukemia is organized as a hierarchy that originates from a primitive hematopoietic cell. *Nature medicine* **3**, 730.
12. Blair, A., Hogge, D., Ailles, L., Lansdorp, P., and Sutherland, H. (1997). Lack of expression of Thy-1 (CD90) on acute myeloid leukemia cells with long-term proliferative ability in vitro and in vivo. *Blood* **89**, 3104-3112.
13. Blair, A., Hogge, D.E., and Sutherland, H.J. (1998). Most Acute Myeloid Leukemia Progenitor Cells With Long-Term Proliferative Ability In Vitro and In Vivo Have the Phenotype CD34+/CD71-/HLA-DR-. *Blood* **92**, 4325-4335. 10.1182/blood.V92.11.4325.
14. Majeti, R., Chao, M.P., Alizadeh, A.A., Pang, W.W., Jaiswal, S., Gibbs, K.D., Jr., van Rooijen, N., and Weissman, I.L. (2009). CD47 is an adverse prognostic factor and therapeutic antibody target on human acute myeloid leukemia stem cells. *Cell* **138**, 286-299. 10.1016/j.cell.2009.05.045.
15. Jan, M., Chao, M.P., Cha, A.C., Alizadeh, A.A., Gentles, A.J., Weissman, I.L., and Majeti, R. (2011). Prospective separation of normal and leukemic stem cells based on differential expression of TIM3, a human acute myeloid leukemia stem cell marker. *Proceedings of the National Academy of Sciences* **108**, 5009-5014. 10.1073/pnas.1100551108.
16. Jan, M., Snyder, T.M., Corces-Zimmerman, M.R., Vyas, P., Weissman, I.L., Quake, S.R., and Majeti, R. (2012). Clonal evolution of preleukemic hematopoietic stem cells precedes human acute myeloid leukemia. *Science translational medicine* **4**, 149ra118-149ra118. 10.1126/scitranslmed.3004315.

17. Dick, J.E., Bhatia, M., Gan, O., Kapp, U., and Wang, J.C. (1997). Assay of human stem cells by repopulation of NOD/SCID mice. *Stem cells (Dayton, Ohio)* **15 Suppl 1**, 199-203; discussion 204-197. 10.1002/stem.5530150826.
18. Goardon, N., Marchi, E., Atzberger, A., Quek, L., Schuh, A., Soneji, S., Woll, P., Mead, A., Alford, K.A., Rout, R., et al. (2011). Coexistence of LMPP-like and GMP-like Leukemia Stem Cells in Acute Myeloid Leukemia. *Cancer Cell* **19**, 138-152. 10.1016/j.ccr.2010.12.012.
19. Sarry, J.E., Murphy, K., Perry, R., Sanchez, P.V., Secreto, A., Keefer, C., Swider, C.R., Strzelecki, A.C., Cavellier, C., Récher, C., et al. (2011). Human acute myelogenous leukemia stem cells are rare and heterogeneous when assayed in NOD/SCID/IL2R γ -deficient mice. *J Clin Invest* **121**, 384-395. 10.1172/jci41495.
20. Kikushige, Y., Shima, T., Takayanagi, S., Urata, S., Miyamoto, T., Iwasaki, H., Takenaka, K., Teshima, T., Tanaka, T., Inagaki, Y., and Akashi, K. (2010). TIM-3 is a promising target to selectively kill acute myeloid leukemia stem cells. *Cell Stem Cell* **7**, 708-717. 10.1016/j.stem.2010.11.014.
21. Jordan, C.T., Upchurch, D., Szilvassy, S.J., Guzman, M.L., Howard, D.S., Pettigrew, A.L., Meyerrose, T., Rossi, R., Grimes, B., Rizzieri, D.A., et al. (2000). The interleukin-3 receptor alpha chain is a unique marker for human acute myelogenous leukemia stem cells. *Leukemia* **14**, 1777-1784. 10.1038/sj.leu.2401903.
22. Jin, L., Lee, E.M., Ramshaw, H.S., Busfield, S.J., Peoppl, A.G., Wilkinson, L., Guthridge, M.A., Thomas, D., Barry, E.F., and Boyd, A. (2009). Monoclonal antibody-mediated targeting of CD123, IL-3 receptor α chain, eliminates human acute myeloid leukemic stem cells. *Cell stem cell* **5**, 31-42.
23. Jin, L., Hope, K.J., Zhai, Q., Smadja-Joffe, F., and Dick, J.E. (2006). Targeting of CD44 eradicates human acute myeloid leukemic stem cells. *Nature medicine* **12**, 1167-1174.
24. Van Rhenen, A., Van Dongen, G.A., Kelder, A., Rombouts, E.J., Feller, N., Moshaver, B., Walsum, M.S.-v., Zweegman, S., Ossenkoppele, G.J., and Jan Schuurhuis, G. (2007). The novel AML stem cell-associated antigen CLL-1 aids in discrimination between normal and leukemic stem cells. *Blood, The Journal of the American Society of Hematology* **110**, 2659-2666.
25. Saito, Y., Kitamura, H., Hijikata, A., Tomizawa-Murasawa, M., Tanaka, S., Takagi, S., Uchida, N., Suzuki, N., Sone, A., and Najima, Y. (2010). Identification of therapeutic targets for quiescent, chemotherapy-resistant human leukemia stem cells. *Science translational medicine* **2**, 17ra19-17ra19.
26. Hosen, N., Park, C.Y., Tatsumi, N., Oji, Y., Sugiyama, H., Gramatzki, M., Krensky, A.M., and Weissman, I.L. (2007). CD96 is a leukemic stem cell-specific marker in human acute myeloid leukemia. *Proceedings of the National Academy of Sciences* **104**, 11008-11013.
27. Pabst, C., Bergeron, A., Lavallée, V.-P., Yeh, J., Gendron, P., Norddahl, G.L., Kros, J., Boivin, I., Deneault, E., and Simard, J. (2016). GPR56 identifies primary human acute myeloid leukemia cells with high repopulating potential in vivo. *Blood, The Journal of the American Society of Hematology* **127**, 2018-2027.
28. Iwasaki, M., Liedtke, M., Gentles, A.J., and Cleary, M.L. (2015). CD93 marks a non-quiescent human leukemia stem cell population and is required for development of MLL-rearranged acute myeloid leukemia. *Cell stem cell* **17**, 412-421.

29. Quek, L., Otto, G.W., Garnett, C., Lhermitte, L., Karamitros, D., Stoilova, B., Lau, I.-J., Doondeea, J., Usukhbayar, B., Kennedy, A., et al. (2016). Genetically distinct leukemic stem cells in human CD34⁺ acute myeloid leukemia are arrested at a hemopoietic precursor-like stage. *The Journal of Experimental Medicine* 213, 1513-1535. 10.1084/jem.20151775.
30. Xie, S.Z., Kaufmann, K.B., Wang, W., Chan-Seng-Yue, M., Gan, O.I., Laurenti, E., Garcia-Prat, L., Takayanagi, S.-i., Ng, S.W.K., Xu, C., et al. (2021). Sphingosine-1-Phosphate Receptor 3 Potentiates Inflammatory Programs in Normal and Leukemia Stem Cells to Promote Differentiation. *Blood Cancer Discovery* 2, 32-53. 10.1158/2643-3230.Bcd-20-0155.
31. Reinisch, A., Thomas, D., Corces, M.R., Zhang, X., Gratzinger, D., Hong, W.J., Schallmoser, K., Strunk, D., and Majeti, R. (2016). A humanized bone marrow ossicle xenotransplantation model enables improved engraftment of healthy and leukemic human hematopoietic cells. *Nature medicine* 22, 812-821. 10.1038/nm.4103.
32. Choo, S., Wolf, C.B., Mack, H.M., Egan, M.J., Kiem, H.P., and Radtke, S. (2024). Choosing the right mouse model: comparison of humanized NSG and NBSGW mice for in vivo HSC gene therapy. *Blood Adv* 8, 916-926. 10.1182/bloodadvances.2023011371.
33. Schmitt, N., Streuer, A., Danner, J., Altmann, E., Riabov, V., Xu, Q., Rapp, F., Weimer, N., Palme, I., Nowak, V., et al. (2022). Direct Comparison of NSG and NBSGW Mice for MDS Patient-Derived Xenograft Models. *Blood* 140, 4009-4010. 10.1182/blood-2022-168709.
34. Tyner, J.W., Tognon, C.E., Bottomly, D., Wilmot, B., Kurtz, S.E., Savage, S.L., Long, N., Schultz, A.R., Traer, E., Abel, M., et al. (2018). Functional genomic landscape of acute myeloid leukaemia. *Nature*. 10.1038/s41586-018-0623-z.
35. Jayavelu, A.K., Wolf, S., Buettner, F., Alexe, G., Häupl, B., Comoglio, F., Schneider, C., Doebele, C., Fuhrmann, D.C., Wagner, S., et al. (2022). The proteogenomic subtypes of acute myeloid leukemia. *Cancer Cell* 40, 301-317.e312. 10.1016/j.ccell.2022.02.006.
36. McKeown, M.R., Corces, M.R., Eaton, M.L., Fiore, C., Lee, E., Lopez, J.T., Chen, M.W., Smith, D., Chan, S.M., Koenig, J.L., et al. (2017). Superenhancer Analysis Defines Novel Epigenomic Subtypes of Non-APL AML, Including an RAR α Dependency Targetable by SY-1425, a Potent and Selective RAR α Agonist. *Cancer Discov* 7, 1136-1153. 10.1158/2159-8290.Cd-17-0399.
37. Ley, T.J., Miller, C., Ding, L., Raphael, B.J., Mungall, A.J., Robertson, A., Hoadley, K., Triche, T.J., Jr., Laird, P.W., Baty, J.D., et al. (2013). Genomic and epigenomic landscapes of adult de novo acute myeloid leukemia. *The New England journal of medicine* 368, 2059-2074. 10.1056/NEJMoa1301689.
38. Metzeler, K.H., Hummel, M., Bloomfield, C.D., Spiekermann, K., Braess, J., Sauerland, M.C., Heinecke, A., Radmacher, M., Marcucci, G., Whitman, S.P., et al. (2008). An 86-probe-set gene-expression signature predicts survival in cytogenetically normal acute myeloid leukemia. *Blood* 112, 4193-4201. 10.1182/blood-2008-02-134411.
39. Tomasson, M.H., Xiang, Z., Walgren, R., Zhao, Y., Kasai, Y., Miner, T., Ries, R.E., Lubman, O., Fremont, D.H., McLellan, M.D., et al. (2008). Somatic mutations and germline sequence variants in the expressed tyrosine kinase genes of patients with de novo acute myeloid leukemia. *Blood* 111, 4797-4808. 10.1182/blood-2007-09-113027.

40. Wouters, B.J., Lowenberg, B., Erpelinck-Verschueren, C.A., van Putten, W.L., Valk, P.J., and Delwel, R. (2009). Double CEBPA mutations, but not single CEBPA mutations, define a subgroup of acute myeloid leukemia with a distinctive gene expression profile that is uniquely associated with a favorable outcome. *Blood* 113, 3088-3091. 10.1182/blood-2008-09-179895.
41. de Jonge, H.J., Valk, P.J., Veeger, N.J., ter Elst, A., den Boer, M.L., Cloos, J., de Haas, V., van den Heuvel-Eibrink, M.M., Kaspers, G.J., Zwaan, C.M., et al. (2010). High VEGFC expression is associated with unique gene expression profiles and predicts adverse prognosis in pediatric and adult acute myeloid leukemia. *Blood* 116, 1747-1754. 10.1182/blood-2010-03-270991.
42. Schimek, M.G., Budinská, E., Kugler, K.G., Švendová, V., Ding, J., and Lin, S. (2015). TopKLists: a comprehensive R package for statistical inference, stochastic aggregation, and visualization of multiple omics ranked lists. *Statistical Applications in Genetics and Molecular Biology* 14, 311-316. doi:10.1515/sagmb-2014-0093.
43. Zeng, A.G.X., Bansal, S., Jin, L., Mitchell, A., Chen, W.C., Abbas, H.A., Chan-Seng-Yue, M., Voisin, V., van Galen, P., Tierens, A., et al. (2022). A cellular hierarchy framework for understanding heterogeneity and predicting drug response in acute myeloid leukemia. *Nature medicine* 28, 1212-1223. 10.1038/s41591-022-01819-x.
44. Nehme, A., Dakik, H., Picou, F., Cheok, M., Preudhomme, C., Dombret, H., Lambert, J., Gyan, E., Pigneux, A., Récher, C., et al. (2020). Horizontal meta-analysis identifies common deregulated genes across AML subgroups providing a robust prognostic signature. *Blood Advances* 4, 5322-5335. 10.1182/bloodadvances.2020002042.
45. Wagner, S., Vadakekolathu, J., Tasian, S.K., Altmann, H., Bornhäuser, M., Pockley, A.G., Ball, G.R., and Rutella, S. (2019). A parsimonious 3-gene signature predicts clinical outcomes in an acute myeloid leukemia multicohort study. *Blood Adv* 3, 1330-1346. 10.1182/bloodadvances.2018030726.
46. Ediriwickrema, A., Nakauchi, Y., Fan, A.C., Köhnke, T., Hu, X., Luca, B.A., Kim, Y., Ramakrishnan, S., Nakamoto, M., Karigane, D., et al. (2025). A single-cell framework identifies functionally and molecularly distinct multipotent progenitors in adult human hematopoiesis. *Cell Reports* 44. 10.1016/j.celrep.2025.116236.
47. Lotfollahi, M., Wolf, F.A., and Theis, F.J. (2019). scGen predicts single-cell perturbation responses. *Nature Methods* 16, 715-721. 10.1038/s41592-019-0494-8.
48. Benaglia, T., Chauveau, D., Hunter, D.R., and Young, D.S. (2009). mixtools: An R Package for Analyzing Mixture Models. *Journal of Statistical Software* 32, 1 - 29. 10.18637/jss.v032.i06.
49. Newman, A.M., Steen, C.B., Liu, C.L., Gentles, A.J., Chaudhuri, A.A., Scherer, F., Khodadoust, M.S., Esfahani, M.S., Luca, B.A., Steiner, D., et al. (2019). Determining cell type abundance and expression from bulk tissues with digital cytometry. *Nature biotechnology*. 10.1038/s41587-019-0114-2.
50. Christopher, M.J., Petti, A.A., Rettig, M.P., Miller, C.A., Chendamarai, E., Duncavage, E.J., Klco, J.M., Helton, N.M., O’Laughlin, M., Fronick, C.C., et al. (2018). Immune Escape of Relapsed AML Cells after Allogeneic Transplantation. *The New England journal of medicine* 379, 2330-2341. 10.1056/NEJMoa1808777.

51. Hackl, H., Steinleitner, K., Lind, K., Hofer, S., Tosic, N., Pavlovic, S., Suvajdzic, N., Sill, H., and Wieser, R. (2015). A gene expression profile associated with relapse of cytogenetically normal acute myeloid leukemia is enriched for leukemia stem cell genes. *Leukemia & lymphoma* 56, 1126-1128. 10.3109/10428194.2014.944523.
52. Li, S., Garrett-Bakelman, F.E., Chung, S.S., Sanders, M.A., Hricik, T., Rapaport, F., Patel, J., Dillon, R., Vijay, P., Brown, A.L., et al. (2016). Distinct evolution and dynamics of epigenetic and genetic heterogeneity in acute myeloid leukemia. *Nature medicine* 22, 792-799. 10.1038/nm.4125.
53. Nuno, K.A., Azizi, A., Koehnke, T., Ediriwickrema, A., Corces, M.R., and Majeti, R. (2021). Chromatin Accessibility Analysis Reveals Epigenetic Evolution Is a Common Mechanism of Relapse in Acute Myeloid Leukemia. *Blood* 138, 677. <https://doi.org/10.1182/blood-2021-148928>.
54. Toffalori, C., Zito, L., Gambacorta, V., Riba, M., Oliveira, G., Bucci, G., Barcella, M., Spinelli, O., Greco, R., Crucitti, L., et al. (2019). Immune signature drives leukemia escape and relapse after hematopoietic cell transplantation. *Nature medicine* 25, 603-611. 10.1038/s41591-019-0400-z.
55. Papaemmanuil, E., Gerstung, M., Bullinger, L., Gaidzik, V.I., Paschka, P., Roberts, N.D., Potter, N.E., Heuser, M., Thol, F., Bolli, N., et al. (2016). Genomic Classification and Prognosis in Acute Myeloid Leukemia. *The New England journal of medicine* 374, 2209-2221. 10.1056/NEJMoa1516192.
56. Tibshirani, R. (1996). Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society. Series B (Methodological)* 58, 267-288.
57. Corces, M.R., Buenrostro, J.D., Wu, B., Greenside, P.G., Chan, S.M., Koenig, J.L., Snyder, M.P., Pritchard, J.K., Kundaje, A., Greenleaf, W.J., et al. (2016). Lineage-specific and single-cell chromatin accessibility charts human hematopoiesis and leukemia evolution. *Nature Genetics* 48, 1193. 10.1038/ng.3646
<https://www.nature.com/articles/ng.3646#supplementary-information>.
58. Majeti, R., Nakauchi, Y., Ediriwickrema, A., Martinez-Krams, D., Zhao, F., Rangavajhula, A., and Karigane, D. (2023). Simplified Intrafemoral Injections Using Live Mice Allow for Continuous Bone Marrow Analysis. *JoVE*, e65874. doi:10.3791/65874.
59. Schuurhuis, G.J., Heuser, M., Freeman, S., Béné, M.-C., Buccisano, F., Cloos, J., Grimwade, D., Haferlach, T., Hills, R.K., Hourigan, C.S., et al. (2018). Minimal/measurable residual disease in AML: a consensus document from the European LeukemiaNet MRD Working Party. *Blood* 131, 1275-1291. 10.1182/blood-2017-09-801498.
60. Majeti, R., Becker, M.W., Tian, Q., Lee, T.-L.M., Yan, X., Liu, R., Chiang, J.-H., Hood, L., Clarke, M.F., and Weissman, I.L. (2009). Dysregulated gene expression networks in human acute myelogenous leukemia stem cells. *Proceedings of the National Academy of Sciences*, pnas. 0900089106.
61. Gentles, A.J., Newman, A.M., Liu, C.L., Bratman, S.V., Feng, W., Kim, D., Nair, V.S., Xu, Y., Khuong, A., Hoang, C.D., et al. (2015). The prognostic landscape of genes and infiltrating immune cells across human cancers. *Nature medicine* 21, 938-945. 10.1038/nm.3909.
62. Palpant, N.J., Wang, Y., Hadland, B., Zaunbrecher, R.J., Redd, M., Jones, D., Pabon, L., Jain, R., Epstein, J., Ruzzo, W.L., et al. (2017). Chromatin and Transcriptional Analysis of

- Mesoderm Progenitor Cells Identifies HOPX as a Regulator of Primitive Hematopoiesis. *Cell Rep* 20, 1597-1608. 10.1016/j.celrep.2017.07.067.
63. Takeda, N., Jain, R., Leboeuf, M.R., Padmanabhan, A., Wang, Q., Li, L., Lu, M.M., Millar, S.E., and Epstein, J.A. (2013). Hopx expression defines a subset of multipotent hair follicle stem cells and a progenitor population primed to give rise to K6+ niche cells. *Development* 140, 1655-1664. 10.1242/dev.093005.
 64. Takeda, N., Jain, R., LeBoeuf, M.R., Wang, Q., Lu, M.M., and Epstein, J.A. (2011). Interconversion between intestinal stem cell populations in distinct niches. *Science (New York, N.Y.)* 334, 1420-1424. 10.1126/science.1213214.
 65. Bourque, J., Kousnetsov, R., and Hawiger, D. (2022). Roles of Hopx in the differentiation and functions of immune cells. *European Journal of Cell Biology* 101, 151242. <https://doi.org/10.1016/j.ejcb.2022.151242>.
 66. Mariotto, A., Pavlova, O., Park, H.-S., Huber, M., and Hohl, D. (2016). HOPX: The Unusual Homeodomain-Containing Protein. *Journal of Investigative Dermatology* 136, 905-911. <https://doi.org/10.1016/j.jid.2016.01.032>.
 67. Chen, F., Kook, H., Milewski, R., Gitler, A.D., Lu, M.M., Li, J., Nazarian, R., Schnepf, R., Jen, K., Biben, C., et al. (2002). Hop Is an Unusual Homeobox Gene that Modulates Cardiac Development. *Cell* 110, 713-723. 10.1016/S0092-8674(02)00932-7.
 68. Trivedi, C.M., Zhu, W., Wang, Q., Jia, C., Kee, H.J., Li, L., Hannenhalli, S., and Epstein, J.A. (2010). Hopx and Hdac2 interact to modulate Gata4 acetylation and embryonic cardiac myocyte proliferation. *Dev Cell* 19, 450-459. 10.1016/j.devcel.2010.08.012.
 69. Jain, R., Li, D., Gupta, M., Manderfield, L.J., Ifkovits, J.L., Wang, Q., Liu, F., Liu, Y., Poleshko, A., Padmanabhan, A., et al. (2015). HEART DEVELOPMENT. Integration of Bmp and Wnt signaling by Hopx specifies commitment of cardiomyoblasts. *Science (New York, N.Y.)* 348, aaa6071. 10.1126/science.aaa6071.
 70. Zhou, X., Crow, A.L., Hartiala, J., Spindler, T.J., Ghazalpour, A., Barsky, L.W., Bennett, B.J., Parks, B.W., Eskin, E., Jain, R., et al. (2015). The Genetic Landscape of Hematopoietic Stem Cell Frequency in Mice. *Stem cell reports* 5, 125-138. 10.1016/j.stemcr.2015.05.008.
 71. Vitali, C., Bassani, C., Chiodoni, C., Fellini, E., Guarnotta, C., Miotti, S., Sangaletti, S., Fuligni, F., De Cecco, L., Piccaluga, P.P., et al. (2015). SOCS2 Controls Proliferation and Stemness of Hematopoietic Cells under Stress Conditions and Its Deregulation Marks Unfavorable Acute Leukemias. *Cancer Research* 75, 2387-2399. 10.1158/0008-5472.Can-14-3625.
 72. Laszlo, G.S., Ries, R.E., Gudgeon, C.J., Harrington, K.H., Alonzo, T.A., Gerbing, R.B., Raimondi, S.C., Hirsch, B.A., Gamis, A.S., Meshinchi, S., and Walter, R.B. (2014). High expression of suppressor of cytokine signaling-2 predicts poor outcome in pediatric acute myeloid leukemia: a report from the Children's Oncology Group. *Leukemia & lymphoma* 55, 2817-2821. 10.3109/10428194.2014.893305.
 73. Zhang, T.Y., Dutta, R., Benard, B., Zhao, F., Yin, R., and Majeti, R. (2020). IL-6 blockade reverses bone marrow failure induced by human acute myeloid leukemia. *Sci Transl Med* 12. 10.1126/scitranslmed.aax5104.
 74. Dorritie, K., McCubrey, J., and Johnson, D. (2014). STAT transcription factors in hematopoiesis and leukemogenesis: opportunities for therapeutic intervention. *Leukemia* 28, 248-257.

75. Sachs, K., Sarver, A.L., Noble-Orcutt, K.E., LaRue, R.S., Antony, M.L., Chang, D., Lee, Y., Navis, C.M., Hillesheim, A.L., Nykaza, I.R., et al. (2020). Single-Cell Gene Expression Analyses Reveal Distinct Self-Renewing and Proliferating Subsets in the Leukemia Stem Cell Compartment in Acute Myeloid Leukemia. *Cancer Research* 80, 458-470. 10.1158/0008-5472.Can-18-2932.
76. Antony, M.L., Chang, D., Noble-Orcutt, K.E., Kay, A., Jensen, J.L., Mohei, H., Myers, C.L., Sachs, K., and Sachs, Z. (2023). CD69 marks a subpopulation of acute myeloid leukemia with enhanced colony forming capacity and a unique signaling activation state. *Leukemia & lymphoma* 64, 1262-1274. 10.1080/10428194.2023.2207698.
77. Zhang, Y., Jiang, S., He, F., Tian, Y., Hu, H., Gao, L., Zhang, L., Chen, A., Hu, Y., Fan, L., et al. (2023). Single-cell transcriptomics reveals multiple chemoresistant properties in leukemic stem and progenitor cells in pediatric AML. *Genome biology* 24, 199. 10.1186/s13059-023-03031-7.
78. Cibrián, D., and Sánchez-Madrid, F. (2017). CD69: from activation marker to metabolic gatekeeper. *Eur J Immunol* 47, 946-953. 10.1002/eji.201646837.
79. Dobin, A., Davis, C.A., Schlesinger, F., Drenkow, J., Zaleski, C., Jha, S., Batut, P., Chaisson, M., and Gingeras, T.R. (2012). STAR: ultrafast universal RNA-seq aligner. *Bioinformatics (Oxford, England)* 29, 15-21. 10.1093/bioinformatics/bts635.
80. Li, B., and Dewey, C.N. (2011). RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome. *BMC Bioinformatics* 12, 323. 10.1186/1471-2105-12-323.
81. Gautier, L., Cope, L., Bolstad, B.M., and Irizarry, R.A. (2004). affy—analysis of Affymetrix GeneChip data at the probe level. *Bioinformatics (Oxford, England)* 20, 307-315. 10.1093/bioinformatics/btg405.
82. Friedman, J.H., Hastie, T., and Tibshirani, R. (2010). Regularization Paths for Generalized Linear Models via Coordinate Descent. *Journal of Statistical Software* 33, 1 - 22. 10.18637/jss.v033.i01.
83. Hemant, I., Udaya, B.K., Eugene, H.B., and Michael, S.L. (2008). Random survival forests. *The Annals of Applied Statistics* 2, 841-860. 10.1214/08-AOAS169.
84. Kuhn, M. (2008). Building Predictive Models in R Using the caret Package. *Journal of Statistical Software* 28, 1 - 26. 10.18637/jss.v028.i05.
85. Erickson, J.R., Mair, F., Bugos, G., Martin, J., Tyznik, A.J., Nakamoto, M., Mortimer, S., and Prlic, M. (2020). AbSeq protocol using the nano-well cartridge-based rhapsody platform to generate protein and transcript expression data on the single-cell level. *STAR protocols* 1, 100092.
86. Stuart, T., Butler, A., Hoffman, P., Hafemeister, C., Papalexi, E., Mauck, W.M., III, Hao, Y., Stoeckius, M., Smibert, P., and Satija, R. (2019). Comprehensive Integration of Single-Cell Data. *Cell* 177, 1888-1902.e1821. 10.1016/j.cell.2019.05.031.
87. Butler, A., Hoffman, P., Smibert, P., Papalexi, E., and Satija, R. (2018). Integrating single-cell transcriptomic data across different conditions, technologies, and species. *Nature biotechnology* 36, 411-420. 10.1038/nbt.4096.
88. Young, M.D., and Behjati, S. (2020). SoupX removes ambient RNA contamination from droplet-based single-cell RNA sequencing data. *GigaScience* 9. 10.1093/gigascience/giaa151.

89. Mulè, M.P., Martins, A.J., and Tsang, J.S. (2022). Normalizing and denoising protein expression data from droplet-based single cell profiling. *Nature communications* 13, 2099. 10.1038/s41467-022-29356-8.
90. McGinnis, C.S., Murrow, L.M., and Gartner, Z.J. (2019). DoubletFinder: doublet detection in single-cell RNA sequencing data using artificial nearest neighbors. *Cell systems* 8, 329-337. e324.
91. Gulati, G.S., Sikandar, S.S., Wesche, D.J., Manjunath, A., Bharadwaj, A., Berger, M.J., Ilagan, F., Kuo, Angera H., Hsieh, R.W., Cai, S., et al. (2020). Single-cell transcriptional diversity is a hallmark of developmental potential. *Science (New York, N.Y.)* 367, 405-411. doi:10.1126/science.aax0249.
92. Burel, J.G., Pomaznoy, M., Lindestam Arlehamn, C.S., Weiskopf, D., da Silva Antunes, R., Jung, Y., Babor, M., Schulten, V., Seumois, G., Greenbaum, J.A., et al. (2019). Circulating T cell-monocyte complexes are markers of immune perturbations. *Elife* 8. 10.7554/eLife.46045.
93. Dixon, P. (2003). VEGAN, A Package of R Functions for Community Ecology. *Journal of Vegetation Science* 14, 927-930.
94. Luca, B.A., Steen, C.B., Matusiak, M., Azizi, A., Varma, S., Zhu, C., Przybyl, J., Espín-Pérez, A., Diehn, M., Alizadeh, A.A., et al. (2021). Atlas of clinically distinct cell states and ecosystems across human solid tumors. *Cell* 184, 5482-5496.e5428. <https://doi.org/10.1016/j.cell.2021.09.014>.
95. Wu, T., Hu, E., Xu, S., Chen, M., Guo, P., Dai, Z., Feng, T., Zhou, L., Tang, W., Zhan, L., et al. (2021). clusterProfiler 4.0: A universal enrichment tool for interpreting omics data. *The Innovation* 2. 10.1016/j.xinn.2021.100141.
96. Hu, Y., and Smyth, G.K. (2009). ELDA: extreme limiting dilution analysis for comparing depleted and enriched populations in stem cell and other assays. *J Immunol Methods* 347, 70-78. 10.1016/j.jim.2009.06.008.